

FLARE: Focused leaf–lesion awareness via explicit relational modeling for field disease severity assessment

Ya Yu^a, Xingcai Wu^a, Qiaoling Wan^a, Yujiao Dan^a, Peijia Yu^{a,*}, Wei Guo^b, Qi Wang^{a,ib,*}

^a State Key Laboratory of Public Big Data, College of Computer Science and Technology, Guizhou University, Guiyang, China

^b Graduate School of Agricultural and Life Sciences, The University of Tokyo, 1-1-1 Midori-cho, Nishitokyo City, Tokyo, Japan

ARTICLE INFO

Dataset link: <http://flare.samlab.cn/>

Keywords:

Plant leaf disease severity assessment
Relational modeling
Deep learning
Contrastive learning

ABSTRACT

Plant leaf disease severity assessment can assist in quantifying the progression of the disease, providing a basis for precise pesticide application, which in turn effectively enhances crop production. Field disease assessment is usually confronted with complex background disturbances, making it difficult for previous methods that primarily rely on hybrid perception to explicitly and independently recognize lesion and leaf regions. The resulting information entanglement introduces bias into severity assessment and ultimately limits the applicability of such methods in real-world scenarios. To address this issue, we propose a focused leaf–lesion awareness via explicit relational modeling (FLARE) for disease severity assessment. The proposed framework comprises two main components: a focused leaf–lesion awareness network (FLLA-Net) and an explicit relational modeling assessment network (ERMA-Net). The FLLA-Net utilizes lesion and leaf masks to guide the network's focus on key regions. The resulting pretrained vision awareness extraction module (VAE-Module) is separately aware of prior visual knowledge of lesions and leaves, providing focused visual features for ERMA-Net. The ERMA-Net shares and freezes the parameters of the VAE-Module to extract hierarchical representations of lesions and leaves explicitly. It then performs a vision relational modeling module (VRM-Module) to simulate the distribution of lesions on leaves. The resulting relational features are refined through a global feature refinement block. A reinforcement-guided contrastive learning strategy is then incorporated to enrich semantic information, followed by a multi-layer perceptron for predicting disease severity levels. During inference, FLARE requires only images, enabling efficient and practical deployment in real-world scenarios. To support real-world applicability, we construct a comprehensive dataset of complex scenes with categories, segmentation masks, severity labels, and corresponding textual annotations. Extensive experimental results demonstrate that our method achieves at least a 3.34% improvement in severity prediction accuracy compared to existing assessment models, confirming the effectiveness and practical potential of FLARE for disease severity assessment. Our code and dataset are available at <http://flare.samlab.cn/>.

1. Introduction

Crops are essential for global food security, and their healthy growth is closely linked to national food strategies and social stability (Omaye et al., 2024). Plant diseases remain a major threat to crop production, causing yield losses of up to 30% in major crops worldwide and resulting in economic damages amounting to hundreds of billions of dollars (Gai and Wang, 2024). Rational pesticide application strategies are essential for maintaining agricultural productivity, yet inaccurate or excessive use not only reduces efficiency but also leads to environmental pollution and increased costs. Accurate assessment of disease severity quantifies the progression of plant diseases and serves as a critical foundation for rational pesticide application. Consequently, disease

severity assessment is of great practical significance for ensuring stable and sustainable grain production.

Traditional plant leaf disease severity typically relies on field surveys conducted by experienced crop protection specialists. These approaches are time-consuming, labor-intensive, and inherently subjective (Mahlein, 2016; Bock et al., 2010), making them unsuitable for large-scale assessments. With the development of machine learning technologies, these methods have been widely applied to assess plant leaf disease severity. Mehra et al. (2016) build a model using multivariate regression and three machine learning methods to predict the risk level of *Stagonospora nodorum* blotch in winter wheat before sowing. Kumari et al. (2025) propose a method for detecting the

* Corresponding authors.

E-mail addresses: yayu_0204@foxmail.com (Y. Yu), wuxingcai_1231@foxmail.com (X. Wu), qlwanggg@foxmail.com (Q. Wan), danyujiao@126.com (Y. Dan), pjyu@gzu.edu.cn (P. Yu), qiawang@gzu.edu.cn (Q. Wang).

<https://doi.org/10.1016/j.compag.2026.111769>

Received 6 November 2025; Received in revised form 28 January 2026; Accepted 11 April 2026

Available online 23 April 2026

0168-1699/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

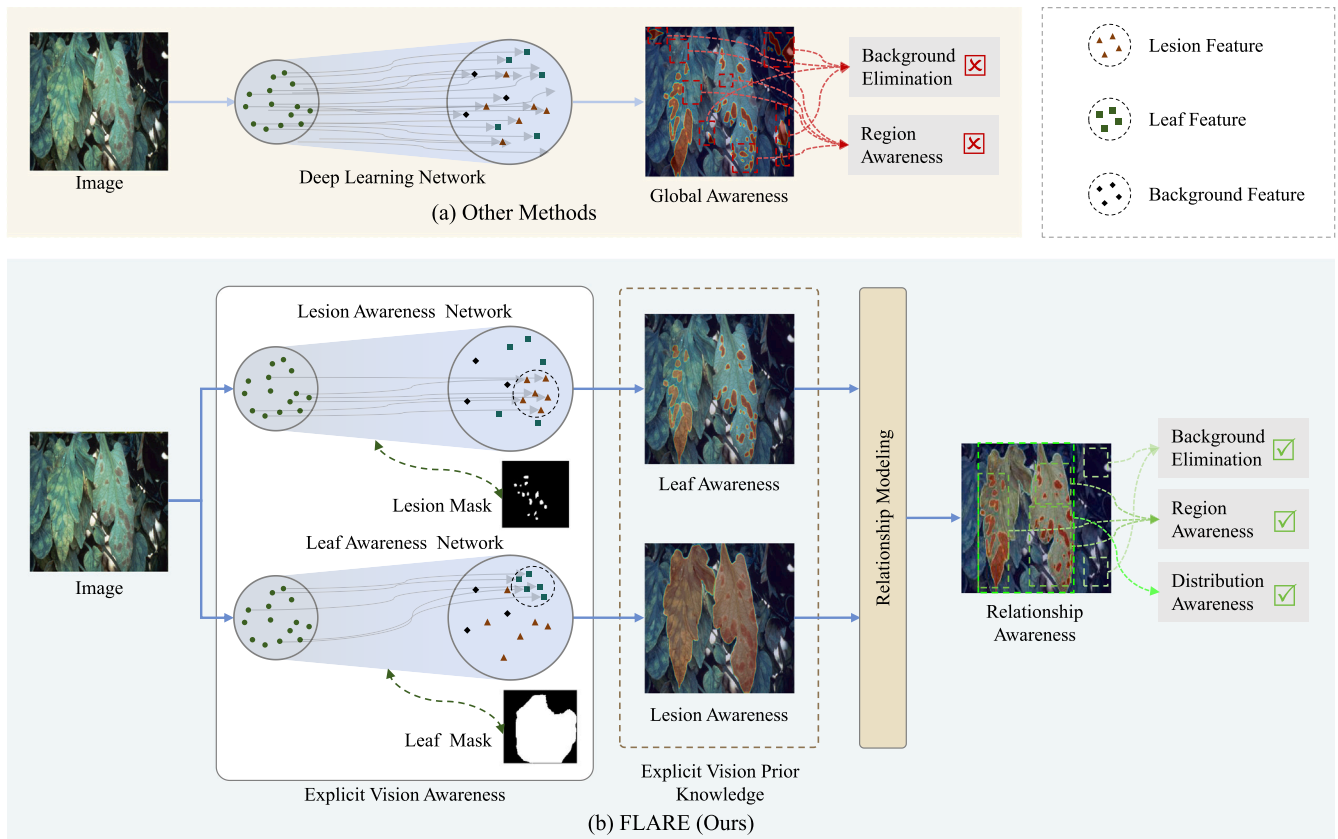


Fig. 1. Comparison of existing deep learning methods and our approach. (a) Global lesion awareness in existing methods mixes background, leaf, and lesion information, lacking the ability for key area perception and background suppression. (b) In contrast, our approach is explicitly aware of lesion and leaf information, effectively suppressing background interference and simulating the distribution of lesions on leaves.

severity of groundnut bud necrosis virus in tomatoes by combining spectral analysis with machine learning. Singh et al. (2023) achieve an accurate assessment of chickpea wilt severity by integrating thermal imaging, visible light imaging, machine learning, and model ensemble techniques. Despite advances in automated inspection, machine learning methods rely on manual feature engineering, leading to limited robustness in real-world applications.

In recent years, deep learning has emerged as a prominent research focus due to its powerful feature learning capabilities (Pacal et al., 2024; Leite et al., 2024). In particular, convolutional neural networks (CNNs) (Haque et al., 2022) and Transformer models (Salamai, 2024) improve the accuracy of disease severity assessment by automatically extracting discriminative features such as lesion texture and color. For example, Bleasdale and Whyatt (2025) fine-tune MobileNetV2 and EfficientNetV2L models in combination with time-series data to analyze classification performance under different lighting conditions and severity levels. Gao et al. (2024) propose an improved YOLOv5s model for the fast and accurate real-time detection of Fusarium head blight in wheat, along with its severity assessment. Yang et al. (2023) design a novel three-branch Swin Transformer classification network, which uses a multi-task learning strategy to classify both the type and severity of plant diseases. However, in real-world field conditions, the complex and heterogeneous backgrounds pose substantial challenges to existing methods. These approaches predominantly rely on hybrid perceptual learning, while the independent attention to lesions or leaf regions is insufficient. This easily results in features entangled with background, leaf, and lesion information, leading to biased severity assessments, as shown in Fig. 1(a). In contrast, human visual awareness tends to focus on lesions and their distribution on the leaf while ignoring irrelevant background information. Motivated by this observation, we introduce explicit visual awareness guided by lesion and leaf masks to obtain their

prior knowledge and then model their relationship, thereby simulating the distribution of lesions on leaves (Fig. 1(b)). Compared to existing methods, our approach integrates background elimination, distribution awareness, and region awareness to achieve more accurate and robust severity estimation.

In detail, we propose a focused leaf-lesion awareness via explicit relational modeling (FLARE) for disease severity assessment. This method is explicitly aware of lesions and leaves, simulating the distribution of lesions on leaves to achieve disease severity assessment that closely resembles human visual perception. Specifically, FLARE consists of two complementary modules: a focused leaf-lesion awareness network (FLA-Net) and an explicit relational modeling assessment network (ERMA-Net). First, FLA-Net incorporates a vision awareness extraction module (VAE-Module) to capture lesion features and extract corresponding leaf features accurately, which is guided by lesion and leaf segmentation masks. Second, ERMA-Net freezes the parameters of the VAE-Module to obtain hierarchical representations of both lesions and leaves, which are subsequently used for relational modeling by the vision relational modeling module (VRM-Module) to simulate the distribution of lesions on leaves. These relational features are first refined by a global feature enhancement block to enhance representation, and further optimized through a multimodal reinforcement-guided contrastive learning strategy to enhance the model's semantic understanding of lesions using textual disease descriptions. The enriched features are subsequently passed through a multi-layer perceptron to predict the final disease severity level. Moreover, a specialized dataset is constructed, incorporating plant and disease categories, segmentation masks, severity, and description, which supports the proposed method's applicability in complex, real-world scenarios. In summary, the contributions of this paper include the following:

Table 1

Statistics of existing plant leaf disease datasets, including the total number of images, number of diseases, number of plant species, coverage, scene complexity, and available annotation types. The symbol “–” represents that the exact quantity is unknown.

Dataset name	Images	Diseases	Plants	Coverage	Scene	Labels
AI Challenger2018 (AI Challenger, 2018)	50,000	27	10	Crop, Fruit	Simple, Complex	Category
PlantVillage (Hughes et al., 2015)	54,309	38	14	Crop, Fruit	Simple	Category, Segment Leaf
PDDD (Dong et al., 2023)	421,133	120	40	Crop, Fruit	Simple, Complex	Category
LWDCD 2020 (Goyal et al., 2021)	12,000	10	1	Wheat Only	Simple, Complex	Category
Leaf Disease Dataset (Sladojevic et al., 2016)	4483	15	5	Fruits Only	Simple, Complex	Category
PhenoBench (Weyler et al., 2024)	2100	–	1	Sugar beet only	Complex	Segment Leaf
CropAndWeed (Steininger et al., 2023)	8034	–	74	Crop, weed	Complex	Category, Segment Leaf
Ours	4,301	68	19	Crop, Fruit	Complex	Comprehensive Labels

Comprehensive Labels: Category, Severity Level, Leaf and Lesion Segmentation Masks, and Image Description

1. We address the problem of complex and heterogeneous field backgrounds by introducing a visual awareness method that suppresses background interference and focuses on key lesion areas for reliable severity estimation.
2. We propose a relational awareness approach that models the relationship between leaves and lesions to simulate their spatial distribution to achieve biologically interpretable severity assessment.
3. We construct a dataset with multi-level annotations for 19 crop species under complex real-world conditions, including categories, segmentation, severity, and descriptions.
4. Our approach improves the accuracy of disease severity assessment by at least 3.34% compared to existing methods.

2. Materials and methods

2.1. Datasets

Deep learning is a data-driven technology that heavily relies on the scale and quality of training data. In the field of plant disease research, limitations in annotation dimensions and scene diversity within existing datasets have significantly hindered the development and application of robust models. Through a systematic review of existing public plant disease datasets, the findings are summarized in Table 1. We identified three main limitations in current public datasets: (1) Existing datasets are predominantly designed for disease classification and provide only categorical labels, as exemplified by AI Challenger 2018 (AI Challenger, 2018), PDDD (Dong et al., 2023), and LWDCD 2020 (Goyal et al., 2021). (2) Although certain datasets, such as PlantVillage (Hughes et al., 2015) and CropAndWeed (Steininger et al., 2023), offer leaf-level segmentation, fine-grained lesion annotation under complex and heterogeneous field conditions remains largely inadequate. (3) Furthermore, most public datasets do not include quantitative annotations of disease severity, which are essential for accurate evaluation and effective disease management.

To address the limitations in existing research, our approach presents a multi-dimensional annotated dataset tailored for complex real-world field scenes, as summarized in Table 1. The dataset annotations cover multiple aspects, including plant and disease classification, leaf and lesion segmentation, disease severity assessment, and image description. It encompasses 19 plant species and 68 types of diseases, with a total of 4301 high-quality images, primarily focusing on crops and fruit-bearing plants. This comprehensive annotation scheme enables in-depth analysis and supports a variety of downstream tasks, thus promoting the advancement of intelligent plant disease diagnosis and management systems.

The dataset construction process is illustrated in Fig. 2 and consists of three main stages: data collection, data cleaning, and data annotation. In the data collection stage, we integrate samples from multiple sources, including PlantVillage (Hughes et al., 2015), PDDD (Dong et al., 2023), AI Challenger 2018 (AI Challenger, 2018), and more (Liu

Table 2

Severity grading standards. The severity of the disease was classified into five levels based on the percentage of lesion area relative to the total leaf area (A).

Grade	Area ratio (A)	Description
Level 1	$0 < A \leq 0.10$	Very low severity
Level 3	$0.10 < A \leq 0.25$	Low severity
Level 5	$0.25 < A \leq 0.40$	Moderate severity
Level 7	$0.40 < A \leq 0.65$	High severity
Level 9	$0.65 < A \leq 1.00$	Very high severity

et al., 2024), covering a diverse range of plant species and disease types. This multi-source integration ensures a comprehensive and representative dataset suitable for complex real-world applications. In the data cleaning stage, we filter valid samples from complex scenarios and remove anomalous data to ensure consistency and reliability. In the data annotation stage, we carry out four types of annotation tasks: category, segmentation, image description, and severity grading. Category labels for plant species and diseases are assigned during the data collection phase. Segmentation masks are initially generated using the Segment Anything Model (SAM) (Kirillov et al., 2023) via the X-AnyLabeling-2.3.5 tool, followed by manual refinement to improve detail accuracy. The disease severity level label is based on the ratio of the lesion area to the leaf area, combined with the disease severity grading standard (National public service platform for standards information, 0000) for quantitative assessment, as shown in Table 2. In addition, image descriptions are manually created to provide detailed textual information that complements the visual data.

2.2. Methods

We propose a focused leaf-lesion awareness via explicit relational modeling (FLARE) for disease severity assessment, as illustrated in Fig. 3. The FLARE method comprises two key components: the focused leaf-lesion awareness network (FLA-Net) and the explicit relational modeling assessment network (ERMA-Net). FLA-Net employs a dual-stream segmentation architecture designed to train the vision awareness extraction module (VAE-Module) via a segmentation-based approach. This module incorporates prior knowledge of vision, enabling it to effectively capture both lesion and leaf feature information. ERMA-Net builds upon this by sharing and freezing the pre-trained VAE-Module to independently extract hierarchical features of lesions and leaves. It further introduces the vision relational modeling module (VRM-Module) to explicitly model the spatial and semantic relationships between lesions and leaf structures. The modeling features are first refined through a global feature enhancement module, then enhanced via a reinforcement-guided contrastive learning strategy to improve the model's semantic understanding of lesion regions. Subsequently, a multi-layer perceptron effectively leverages these refined features to accurately predict the severity level. Through the collaborative optimization between FLA-Net and ERMA-Net, the proposed method provides a robust solution for plant leaf disease severity assessment. It effectively addresses challenges in complex field conditions, such

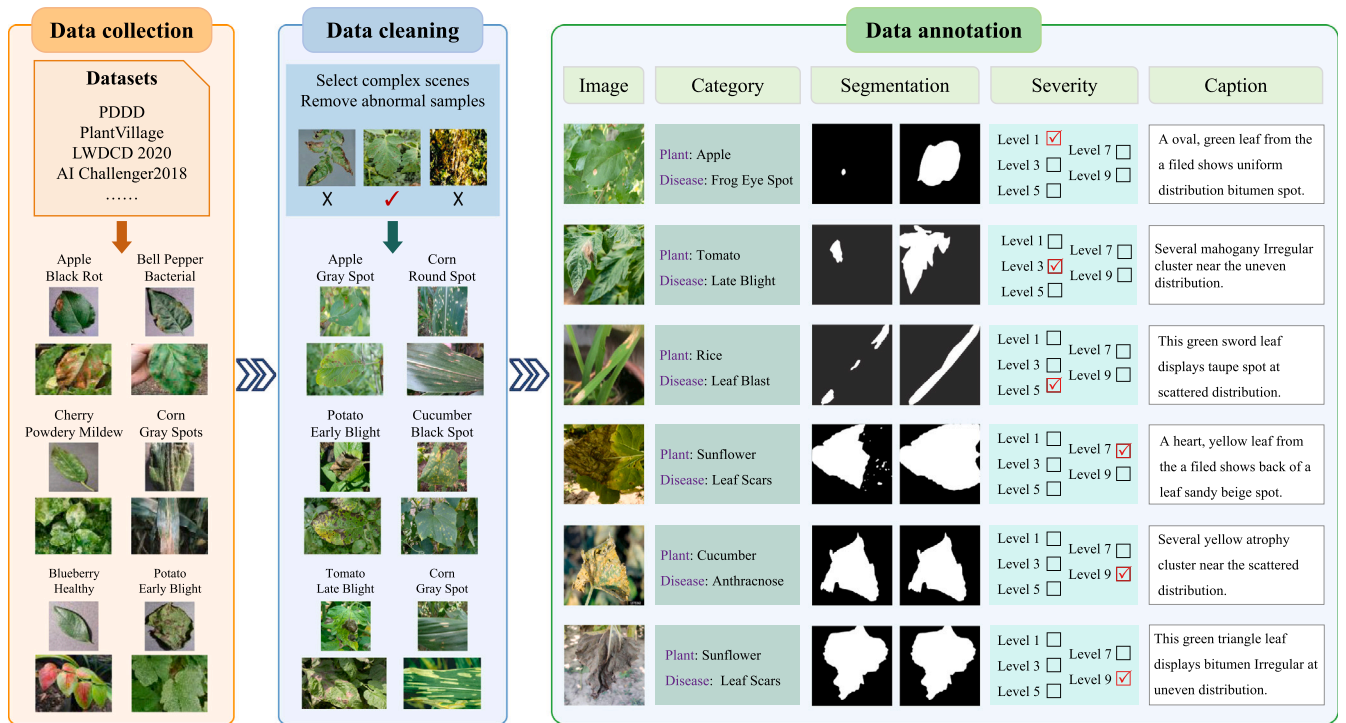


Fig. 2. Workflow of dataset construction, consisting of three main stages: data collection, data cleaning, and data annotation.

as background interference and variable lesion distribution, advancing intelligent disease monitoring and management.

2.2.1. Focused leaf-lesion awareness network (FLLA-Net)

Plant leaf disease phenotypes are diverse and complex, making accurate severity assessment challenging. In complex field scenarios, reliable evaluation requires the model to be effectively and independently aware of both lesions and leaves. To address this, we propose a focused leaf-lesion awareness network (FLLA-Net), as illustrated in Fig. 3(a). The network employs the vision awareness extraction module (VAE-Module), which consists of a lesion awareness unit and a leaf awareness unit, to perform awareness modeling for both lesions and leaves. The leaf awareness unit, based on a segmentation backbone (e.g., UNet++ (Zhou et al., 2019)), is responsible for extracting spatial features associated with the leaf structure, followed by a segmentation head that outputs binary masks for the leaves. The lesion awareness unit uses the same segmentation backbone (Zhou et al., 2019) integrated with a fast Fourier attention module (FFA-Module) to enhance lesion feature extraction by incorporating frequency-domain information. An assist space constraint branch is introduced during the decoding stage to provide spatial constraints. Each branch is followed by a segmentation head that generates binary masks for lesions and leaves. By effectively distinguishing lesions, leaves, and background through the segmentation strategy, FLLA-Net fully leverages the advantages of the VAE-Module to achieve precise independent awareness of both lesions and leaves.

Vision Awareness Extraction Module (VAE-Module). Explicitly distinguishing lesions from leaves is crucial, as it aligns with human visual awareness mechanisms. To this end, we propose the vision awareness extraction module (VAE-Module), which consists of a leaf awareness unit and a lesion awareness unit, enabling explicit and disentangled feature extraction for both components. For leaf awareness, the backbone segmentation network serves as the leaf awareness unit, where nested skip connections and dense convolutions enable multi-scale feature fusion, allowing accurate extraction of leaf morphology while preserving fine edge details. For the lesion awareness task, we construct two decoder branches based on shared encoder features: a

leaf assist space constraint branch decoder and a lesion decoder. The leaf assist branch provides spatial priors that constrain the lesion awareness process, thereby enhancing the model's localization capability. Given that lesions often exhibit complex and subtle details, precise modeling of fine-grained features is essential. To this end, we introduce an innovative fast Fourier attention module (FFA-Module) into the backbone architecture to improve the representation of lesion features. Specifically, the FFA-Module injects the frequency details of the low-level information X_e from the first encoder layer into the high-level semantic features X_d from the third decoder layer. The fused output is then skip-connected and concatenated with features X_4 from the final decoder layer to generate the final output feature map. This process is defined as follows:

$$X_{all} = \text{Concat}(FFA(X_e, X_d), X_4), \quad (1)$$

where $\text{Concat}(\cdot)$ represents the connection in the channel dimension, and FFA means the operation of the fast Fourier attention module.

Fast Fourier Attention Module (FFA-Module). In assessing disease severity in plant leaves, detailed lesion information plays a critical role, as it accurately reflects both the spread and morphological characteristics of the disease. To fully capture such details, we introduce a frequency-domain analysis approach, in which high-frequency components correspond to local details such as edges and textures, while low-frequency components represent the global structure and background information of the image. Based on this, we integrate the Fourier transform with an attention mechanism and propose the fast Fourier attention (FFA) module, as shown in Fig. 4. Specifically, the low-level features X_e and high-level features X_d are transformed into the frequency domain using fast Fourier transform (FFT), where attention weights are calculated to highlight lesion edges and texture details. These weights are then converted back to the spatial domain with inverse fast Fourier transform (IFFT) and fused with X_d to enhance lesion details. The process is repeated to further refine the lesion representation and improve awareness performance. The corresponding

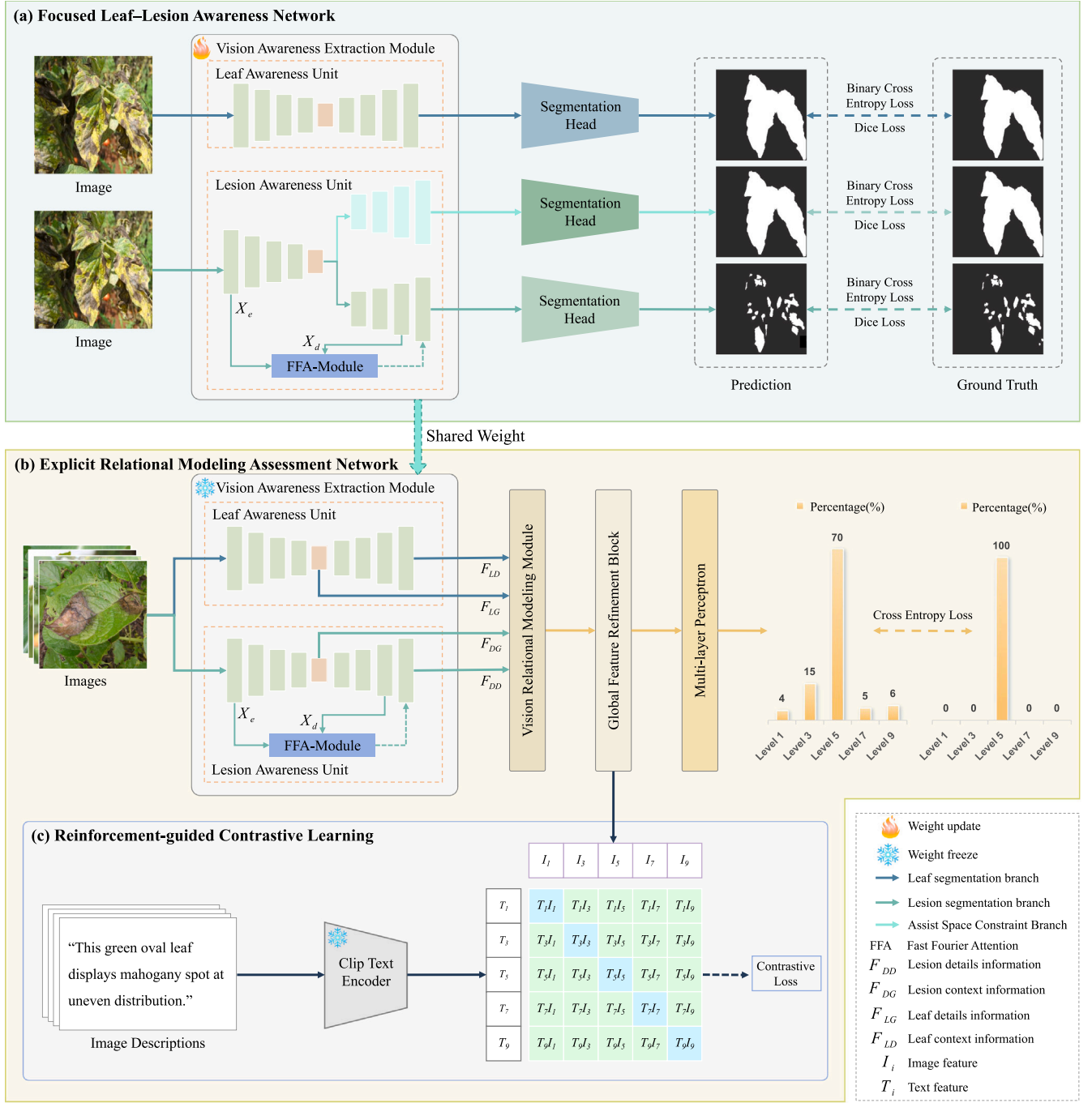


Fig. 3. Framework of our approach. The approach consists of two key components: the focused leaf-lesion awareness network (FLA-Net) and the explicit relational modeling assessment network (ERMA-Net).

mathematical formulation is as follows:

$$\begin{aligned}
 FFA(X_e, X_d) &= FFT_A(FFT_A(X_e, X_d), X_e), \\
 FFT_A(X_e, X_d) &= IFFT(AW(X_e, X_d))X_d, \\
 AW(X_e, X_d) &= Softmax\left(\frac{FFT(X_e)(FFT(X_d))^T}{\sqrt{d}}\right),
 \end{aligned} \tag{2}$$

where FFT denotes the fast Fourier transform, $IFFT$ denotes the inverse fast Fourier transform, X_e represents the feature map from layer 1 of the encoder, X_d represents the feature map from layer 3 of the decoder, and $\sqrt{d}$ is the scaling factor.

Object Function. To enhance the model's ability to localize lesion regions, we adopt a combined loss function composed of binary cross-entropy (BCE) and Dice loss. These two loss functions work jointly to optimize the segmentation accuracy and improve the awareness performance. The binary cross-entropy loss $\mathcal{L}_{bce}$ evaluates the model's performance on pixel-wise binary classification, thereby guiding the network to accurately distinguish lesion or leaf areas at the pixel level. It is formulated as:

$$\mathcal{L}_{bce}(P, Y) = -\frac{1}{N} \sum_{i=1}^N (Y_i \log_2 P_i + (1 - Y_i) \log_2 (1 - P_i)), \tag{3}$$

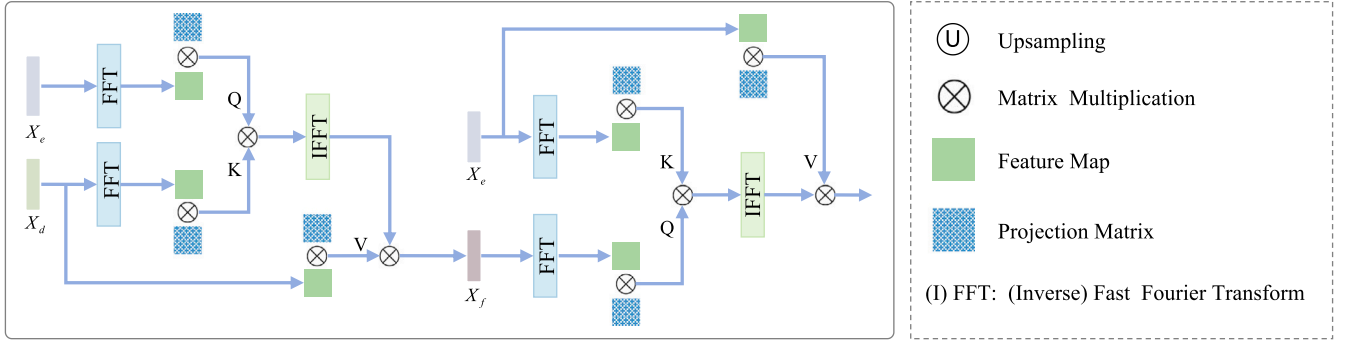


Fig. 4. Framework of the fast Fourier attention module (FFA-Module). It leverages a cascaded FFT attention mechanism to capture detailed information of the lesion by injecting frequency-domain information.

where P denotes the predicted probability map, Y is the ground truth label map, i indexes the pixels, and N is the total number of pixels. Specifically, P_i and Y_i indicate the predicted value and ground truth label of the i th pixel, respectively.

To address the issue of class imbalance, especially in cases where lesion regions occupy a small portion of the image, Dice loss $\mathcal{L}_{dice}$ is introduced, emphasizing region-level similarity and defined as:

$$\mathcal{L}_{dice}(P, Y) = 1 - \frac{2 \sum_{i=1}^N P_i Y_i + \epsilon}{\sum_{i=1}^N (P_i)^2 + \sum_{i=1}^N (Y_i)^2 + \epsilon}, \quad (4)$$

where ϵ is a small constant added for numerical stability to prevent division by zero. In summary, we combine the strengths of both pixel-wise accuracy and regional overlap evaluation by integrating binary cross-entropy and Dice loss into a unified loss function. This composite loss function is expressed as:

$$\mathcal{L}_{all} = \mathcal{L}_{bce}(P, Y) + \mathcal{L}_{dice}(P, Y), \quad (5)$$

For the leaf segmentation branch, the objective is $\mathcal{L}_{all}$. For the lesion segmentation branch, an additional leaf-assisted space constraint is applied, so its loss combines the lesion segmentation loss and the related leaf segmentation loss. The overall loss is defined as:

$$\mathcal{L}_{all_{lesion}} = \mathcal{L}_{all}(lesion) + \mathcal{L}_{all}(leaf). \quad (6)$$

where $\mathcal{L}_{all}(lesion)$ denotes the loss associated with lesion segmentation, while $\mathcal{L}_{all}(leaf)$ refers to the loss related to leaf segmentation.

2.2.2. Explicit relational modeling assessment network (ERMA-Net)

The assessment of plant leaf disease severity relies on the relationship modeling between lesion semantic features and leaf structural information, as the distribution of the lesion area directly reflects the progression of the disease. Although the pre-trained FLLA-Net effectively extracts both types of features, the lack of relationship modeling hinders a comprehensive characterization of disease severity. To address this, we propose the explicit relational modeling assessment network (ERMA-Net), as illustrated in Fig. 3(b), which consists of the following main modules: the pre-trained VAE-Module, the vision relational modeling module (VRM-Module), the global feature refinement block, the reinforcement-guided contrastive learning, and a multi-layer perceptron. First, FLLA-Net shares and freezes the VAE-Module, which is responsible for extracting detailed and contextual features of lesions and leaves. Second, the VRM-Module employs a hierarchical modeling strategy based on a cross-attention mechanism to effectively handle the interdependencies between lesions and leaves. These fused features are subsequently refined and enhanced through the global feature refinement block. Next, the reinforcement-guided contrastive learning module introduces a reinforcement-guided multimodal contrastive learning strategy to further optimize feature representations. Finally, the contrastively learned features are input into a multi-layer perceptron, which outputs the probability for each disease severity level. The

level with the highest probability is selected as the final assessment result.

Vision Awareness Extraction Module (VAE-Module). Accurate disease severity assessment in field scenes requires precise awareness of the detailed and contextual features of both lesions and leaves. Detailed information captures fine-grained characteristics like edges, texture, and morphology, which are crucial for identifying early-stage or mild infections. Contextual information, on the other hand, describes the relationships between lesions and surrounding tissues, aiding in a comprehensive understanding of disease distribution and progression. The VAE-Module, derived from the pre-trained FLLA-Net, is capable of directly extracting these two types of information from diseased images. Specifically, the feature map F_{DG} from the final layer of the lesion awareness unit encoder extracts the contextual semantic information of the lesions, while F_{DD} from the lesion branch decoder captures the fine-grained details of the lesion. Similarly, the feature map F_{LG} from the final layer of the leaf awareness unit encoder provides global structural information of the leaves, and F_{LD} from the decoder extracts their detailed features. Through the VAE-Module, the model is thus able to comprehensively acquire multi-level semantic representations of both lesions and leaves, forming a robust foundation for subsequent feature fusion and severity assessment.

Vision Relational Modeling Module (VRM-Module). To effectively assess the severity of plant disease, the model needs not only to accurately capture the fine-grained features of lesions and leaves but also to understand the complex semantic and structural relationships between them. This is because disease severity is highly correlated with how lesions are distributed over the leaf surface, including their relative scale, spatial extent, and coverage proportion, rather than lesion appearance alone. Therefore, designing a coordinated and efficient modeling strategy to model multi-level information representations is particularly important. To this end, we propose a vision relational modeling module (VRM-Module) to simulate the distribution of lesions on leaves. Specifically, we explore three feature modeling stages to explore the impact of different information sets on performance, as illustrated in Fig. 5, and described as follows:

(1) Detail-focused Modeling (DF Modeling): In the first stage, we focus on capturing the relationships between lesion detailed features and leaf detailed features through a cross-attention mechanism. Cross-attention is first computed between F_{DD} and F_{LD} to extract the correlations between lesion and leaf detail features. Subsequently, a self-attention operation is applied to F_{LD} to enhance its internal structural representations. The resulting features from both attention mechanisms

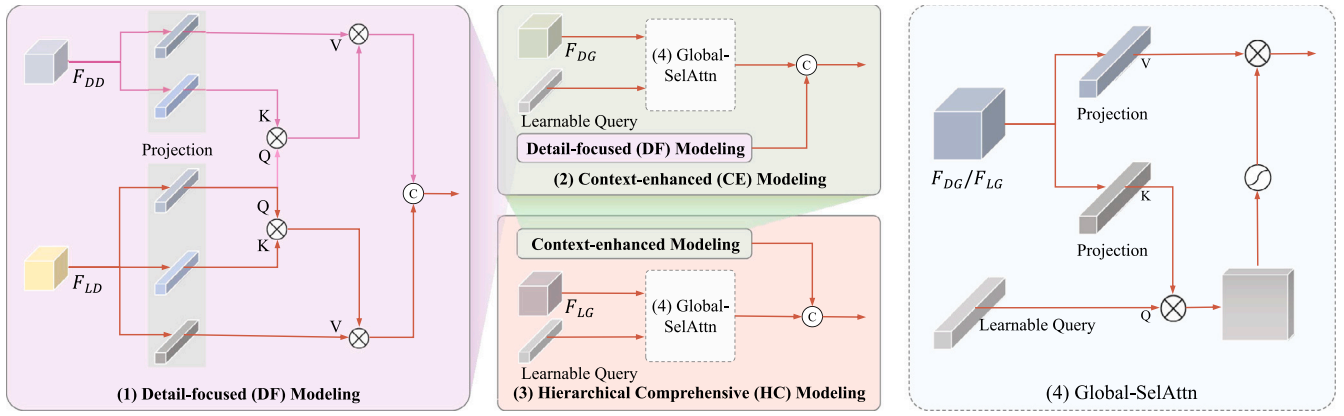


Fig. 5. Framework of the vision relational modeling module (VRM-Module). The VRM-Module incorporates three relation modeling strategies: detail-focused (DF) modeling, context-enhanced (CE) modeling, and hierarchical comprehensive (HC) modeling.

are fused via concatenation. The computation process is as follows:

$$\begin{aligned}
 F_{CSA} &= \text{Concat}(CA(F'_{DD}, F'_{LD}), SA(F'_{LD})), \\
 CA(F'_{DD}, F'_{LD}) &= \text{Softmax}\left(\frac{F'_{LD}(F'_{DD})^T}{\sqrt{d}}\right) F'_{DD}, \\
 SA(F'_{LD}) &= \text{Softmax}\left(\frac{F'_{LD}(F'_{LD})^T}{\sqrt{d}}\right) F'_{LD}, \\
 F'_{DD}, F'_{LD} &= \text{Projection}(F_{DD}), \text{Projection}(F_{LD}),
 \end{aligned} \quad (7)$$

where $\sqrt{d}$ denotes the scaling factor. $\text{Projection}(F_{DD})$ and $\text{Projection}(F_{LD})$ project F_{DD} and F_{LD} into the same feature space, respectively. Moreover, CA and SA denote the cross-attention and self-attention operations, respectively.

(2) Context-enhanced Modeling (CE Modeling): In the second stage, the DF modeling is extended by incorporating the global contextual information F_{DG} of the lesion region, to further enhance the feature representation capability. Specifically, a global self-attention mechanism is applied to F_{DG} , introducing a learnable query vector Q to adaptively extract key contextual information from F_{DG} , thereby improving the model's ability to perceive the overall structure of the lesion. The computation process is as follows:

$$\begin{aligned}
 F_{CGS} &= \text{Concat}(F_{CSA}, \text{Global}(F'_{DG})), \\
 \text{Global}(F'_{DG}) &= \text{Softmax}\left(\frac{Q(F'_{DG})^T}{\sqrt{d}}\right) F'_{DG}, \\
 F'_{DG} &= \text{Projection}(F_{DG}),
 \end{aligned} \quad (8)$$

where F_{CSA} performs the output of the DF Modeling.

(3) Hierarchical Comprehensive Modeling (HC Modeling): The third stage further extends CE Modeling by incorporating global contextual information from the leaf features F_{LG} . With the help of a learnable query vector Q' , it identifies key regions within the leaf context features, enabling multi-level interactions between global information. This approach enhances the model's ability to capture the relationship between leaf morphology and lesion distribution. The computation process is as follows:

$$\begin{aligned}
 F_{MCS} &= \text{Concat}(F_{CGS}, \text{Global}(F'_{LG})), \\
 \text{Global}(F'_{LG}) &= \text{Softmax}\left(\frac{Q'(F'_{LG})^T}{\sqrt{d}}\right) F'_{LG}, \\
 F'_{LG} &= \text{Projection}(F_{LG}),
 \end{aligned} \quad (9)$$

where F_{CGS} performs the output of the CE Modeling.

HC Modeling achieves the best performance by effectively integrating fine-grained lesion and leaf features with global contextual information through multi-level interactive modeling. This enhances the understanding of distributions and improves assessment accuracy.

Global Feature Refinement Block. To improve feature representation and align with text for contrastive learning, the feature map must be refined to preserve semantic consistency and robustly integrate lesion and leaf details. Since the CLIP text encoder produces a 512-dimensional embedding, the image features are projected into the same 512-dimensional space. Specifically, the global feature refinement block first applies a convolutional layer, followed by batch normalization and ReLU activation, to adjust the channel dimension of the fused feature map to 512 and normalize the feature distribution. It then incorporates a residual block to facilitate gradient flow and enhance semantic representation capacity. Finally, adaptive average pooling is used to generate a compact global representation by aggregating spatial information.

Given the modeling feature map F , the block processes it as follows:

$$\begin{aligned}
 F_{out} &= \text{AvgPool}(F''), \\
 F'' &= \text{ResBlock}(F'), \\
 F' &= \text{ReLU}(\text{BN}(\text{Conv}(F_{MCS}))),
 \end{aligned} \quad (10)$$

where F_{MCS} represents the fusion feature from VRM-Module, Conv is a 3×3 convolution, BN denotes batch normalization, ResBlock is a standard residual block, and AvgPool refers to adaptive average pooling to obtain a global feature vector.

Reinforcement-Guided Contrastive Learning. In real-world field conditions, lesion morphology exhibits significant similarity, making it difficult to visually distinguish between diseases of different severity levels. Image descriptions, as high-level semantic priors, provide valuable clues such as color, shape, and spatial distribution of lesions, which help enhance the model's semantic understanding. Therefore, we propose reinforcement-guided contrastive learning, as illustrated in Fig. 3(c). This module leverages a multimodal contrastive learning framework to guide the model in effectively capturing the semantic alignment between images and their corresponding severity-level descriptions. Specifically, reinforcement-guided contrastive learning first constructs image-text pairs for contrastive learning. An input image I_i is passed through the global feature refinement block to extract visual features f_i , while its corresponding textual description T_i is encoded using a CLIP (Radford et al., 2021) text encoder to obtain textual features t_i . This forms a positive sample pair $\langle f_i^+, t_i^+ \rangle$. For negative sample construction, a textual description T_j from an image I_j with different severity levels from I_i is randomly selected and encoded to form the negative pair $\langle f_i^-, t_j^- \rangle$. To enhance the model's ability to learn from hard-to-distinguish samples, we introduce a reinforcement-guided weighting mechanism that dynamically adjusts sample importance based on prediction confidence as a reward signal. For positive samples, a higher predicted probability of the ground-truth class indicates stronger discriminative power and is assigned a larger weight p_i^+ .

For negative samples, a higher confidence in incorrect classes suggests greater ambiguity, and thus a larger weight p_j^- is assigned to emphasize the separation of semantically confusing pairs. The final contrastive loss function is defined as:

$$\mathcal{L}_{rgc} = - \sum_i \log \left(\frac{Record_{pos_i}}{Record_{pos_i} + Record_{neg_i}} \right),$$

$$Record_{pos} = \langle t_i^+, f_i^+ \rangle \cdot p_i^+, \quad (11)$$

$$Record_{neg} = \sum_j \langle t_j^-, f_j^- \rangle \cdot p_j^-,$$

where $\langle \cdot \rangle$ is used to compute the similarity between image and text modalities, p_i^+ and p_j^- represent the weighted similarities of positive and negative sample pairs.

Object Function. To optimize ERMA-Net, two types of loss functions are employed: an auxiliary loss derived from the contrastive learning strategy and a primary classification loss. Plant leaf disease severity is divided into five levels, corresponding to a multi-class classification task. We apply cross-entropy loss, which is defined as follows:

$$\mathcal{L}_{cls} = - \sum_{i=1}^N y_i \log(\hat{y}_i), \quad (12)$$

where N is the number of classes, y_i is the true label, and $\hat{y}_i$ is the probability distribution predicted by the model. The total loss function is therefore defined as:

$$\mathcal{L}_{all} = a\mathcal{L}_{rgc} + b\mathcal{L}_{cls}, \quad (13)$$

where the coefficients a and b are weighting parameters used to balance the contributions of the two loss components. Experimental results demonstrate that the best performance is achieved when $a = 0.1$ and $b = 1$.

2.3. Stagewise training and inference

Our method consists of two sub-networks: FLLA-Net and ERMA-Net. FLLA-Net is responsible for capturing the feature information of lesions and leaves, while ERMA-Net models the relationship between them and estimates the severity level of plant disease based on the relational features. However, training these two sub-networks simultaneously may lead to learning competition, which compromises the accurate awareness of lesion and leaf features, ultimately degrading the overall performance. To address this issue, we design a two-stage training strategy, as detailed in Algorithm 1: ① Stage 1: Pre-training of FLLA-Net. In this stage, the network is trained for a segmentation task using input images and their corresponding segmentation masks. This process enables the VAE-Module within FLLA-Net to be effectively aware of lesion and leaf features. Upon completion of training, the learned weights of the VAE-Module are preserved for subsequent use. ② Stage 2: Training of ERMA-Net. During this stage, the pretrained VAE-Module is frozen and used solely as a feature extractor for lesion and leaf information. The remaining components of ERMA-Net are trained from scratch using image data, textual descriptions, and severity-level annotations, allowing the network to learn the associations among multimodal inputs. ③ Testing stage: After completing the above two stages, test images are fed into ERMA-Net to perform severity estimation. This phase does not require textual input; inference is conducted solely based on images, with minimal input requirements and reduced computational overhead, making it practical for real-world field applications. The predicted severity levels are then compared with ground-truth labels to evaluate model performance. By adopting this two-stage training strategy, we effectively mitigate the learning conflict between sub-networks, fully exploit the awareness and reasoning capabilities of both, and significantly improve the accuracy of disease severity assessment.

Algorithm 1 Two-Stage Training and Inference Strategy for Disease Severity Assessment

Input: Training images $\mathcal{I}_{train}$, segmentation masks $\mathcal{M}_{train}$, textual descriptions $\mathcal{T}_{train}$, severity labels $\mathcal{Y}_{train}$, test images $\mathcal{I}_{test}$, test labels $\mathcal{Y}_{test}$

Output: Predicted severity levels $\hat{\mathcal{Y}}_{test}$

- 1: **Stage 1: Pre-training of FLLA-Net**
- 2: Initialize FLLA-Net parameters θ_D
- 3: **for** epoch = 1 to E_1 **do**
- 4: **for** each $(I, M) \in (\mathcal{I}_{train}, \mathcal{M}_{train})$ **do**
- 5: $F_{seg} \leftarrow \text{FLLA-Net}(I; \theta_D)$
- 6: $\mathcal{L}_{seg} \leftarrow \text{SegmentationLoss}(F_{seg}, M)$
- 7: $\theta_D \leftarrow \theta_D - \eta \nabla \mathcal{L}_{seg}$
- 8: **end for**
- 9: **end for**
- 10: Save weights of VAE-Module $\theta_{LLA} \subset \theta_D$
- 11: **Stage 2: Training of ERMA-Net**
- 12: Initialize ERMA-Net parameters θ_L
- 13: Load θ_{LLA} and freeze gradients
- 14: **for** epoch = 1 to E_2 **do**
- 15: **for** each $(I, T, Y) \in (\mathcal{I}_{train}, \mathcal{T}_{train}, \mathcal{Y}_{train})$ **do**
- 16: $F_{feat} \leftarrow \text{VAE-Module}(I; \theta_{LLA})$
- 17: $F_{rel} \leftarrow \text{ERMA-Net}(F_{feat}, T; \theta_L)$
- 18: $\hat{Y} \leftarrow \text{Classifier}(F_{rel}; \theta_L)$
- 19: $\mathcal{L}_{cls} \leftarrow \text{ClassificationLoss}(\hat{Y}, Y)$
- 20: $\theta_L \leftarrow \theta_L - \eta \nabla \mathcal{L}_{cls}$
- 21: **end for**
- 22: **end for**
- 23: **Stage 3: Inference**
- 24: **for** each $I \in \mathcal{I}_{test}$ **do**
- 25: $F_{feat} \leftarrow \text{VAE-Module}(I; \theta_{LLA})$
- 26: $\hat{Y} \leftarrow \text{ERMA-Net}(F_{feat}; \theta_L)$
- 27: Compare $\hat{Y}$ with $\mathcal{Y}_{test}$ for evaluation
- 28: **end for**
- 29: **return** $\hat{\mathcal{Y}}_{test}$

2.4. Experimental setting

For the experimental settings, all model training and evaluation procedures are conducted using the PyTorch deep learning framework. The environment is configured as follows: Python version 3.8.20, PyTorch version 2.3.0, torchvision version 0.18.0, torchaudio version 2.3.0, and CUDA version 11.8. The experiments are performed on an NVIDIA L20 GPU. For the training process, FLLA-Net is optimized using the Stochastic Gradient Descent (SGD) optimizer with an initial learning rate of 0.1, and is trained for 100 epochs to ensure sufficient convergence on the training set. In contrast, ERMA-Net is trained using the Adaptive Moment Estimation (Adam) optimizer, with a learning rate set to 1×10^{-4} , and the training is carried out for more than 100 epochs to enhance its modeling capacity and generalization ability on complex tasks. We split the dataset at a ratio of 8:2 for each severity level, using 80% of the samples from each level as the training set for model training and parameter optimization, and reserving the remaining 20% as the test set to assess the model's performance. The subsets from all severity levels are then combined to form the final training and test sets, ensuring balanced representation across severity levels and improving the reliability of the evaluation. During the preprocessing stage, all input images are uniformly resized to a resolution of 224×224 pixels to match the input size requirements of the backbone networks.

2.5. Evaluation methods

Since the two networks are designed to perform distinct tasks, we select specific evaluation metrics to assess their performance. For

FLLA-Net, we utilize several commonly adopted evaluation metrics, including precision, recall, F1 score, and intersection over union (IoU). In contrast, the performance of ERMA-Net is primarily evaluated using accuracy. First, we define the key terms as follows: True Positive (TP) refers to the number of positive samples correctly identified by the model, indicating the accurate prediction of positive instances. False Positive (FP) denotes the number of negative samples incorrectly classified as positive. True Negative (TN) corresponds to the number of negative samples correctly identified by the model, representing accurate negative predictions. False Negative (FN) reflects the number of positive samples that the model incorrectly predicts as negative. The following provides a detailed description of each evaluation metric:

Precision. Precision quantifies the proportion of correctly predicted positive samples out of all the predicted positive samples. It is defined as:

$$Precision = \frac{TP}{TP + FP}, \quad (14)$$

Recall. Recall measures the proportion of actual positive samples that are correctly identified by the model, and is defined as:

$$Recall = \frac{TP}{TP + FN}, \quad (15)$$

F1 Score. The F1 score is the harmonic mean of Precision and Recall, providing a balanced measure of the model's ability to predict positive samples. It is calculated as:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}, \quad (16)$$

IoU. The Intersection over Union (IoU) is a metric that quantifies the overlap between the predicted results and the true labels, defined as:

$$IoU = \frac{TP}{TP + FN + FP}, \quad (17)$$

Accuracy. Accuracy is a basic metric for evaluating the performance of an assessment model, indicating how well the model classifies samples. It is calculated as:

$$Accuracy = \frac{T_a}{T_n} \times 100\%. \quad (18)$$

where T_a denotes the number of samples correctly classified by the model, and T_n is the total number of samples in the test set.

By using these distinct evaluation metrics, we can separately assess the performance of the two networks for their respective tasks, ensuring that the model achieves high precision and stability in both feature-aware and disease severity assessment.

2.6. Other compared methods

2.6.1. Other assessment segmentation models

To comprehensively evaluate the performance of the proposed FLARE framework, we compare it with a range of representative deep learning architectures, including convolutional neural networks and Transformer-based or hybrid architectures:

Convolutional neural networks. Several CNN-based architectures are included as strong baselines for visual representation learning. ResNet18 and ResNet34 (He et al., 2016) employ residual connections to alleviate gradient vanishing and facilitate deep feature extraction. DenseNet169 and DenseNet264 (Steen et al., 2018) improve feature propagation through dense connectivity, enhancing information flow and reuse. RegNet (Radosavovic et al., 2020) explores a systematic design space that balances accuracy and efficiency across different scales. ConvNeXt (Liu et al., 2022) achieves improved performance through large-kernel convolutions and normalization refinements. EfficientNet (Tan, 2019) and EfficientNetV2 (Tan and Le, 2021) utilize compound scaling strategies to jointly optimize network depth, width, and resolution, providing strong generalization on diverse visual tasks.

In addition, lightweight networks such as ShuffleNet (Ma et al., 2018), MobileNetV2 (Sandler et al., 2018), and MobileNetV3 (Howard et al., 2019) demonstrate efficient architectural designs that further reflect the evolution of convolution-based models.

Transformer-based architectures. To further benchmark FLARE against models capable of global dependency modeling, we also include transformer-based and hybrid architectures. ViT (Dosovitskiy et al., 2020) introduces the self-attention mechanism into visual recognition, enabling long-range contextual reasoning. RepViT (Wang et al., 2024) employs structural re-parameterization to transform vision transformer designs into a pure CNN architecture, thereby enhancing both feature expressiveness and computational efficiency. SBCFormer (Lu et al., 2024) combines the advantages of CNNs and Transformers and leverages structural re-parameterization to achieve high accuracy while significantly improving inference speed. PDiscoNet (van der Klis et al., 2023) is a partial distillation separation technology that enhances the model's feature representation capabilities without introducing additional parameters.

2.6.2. Segmentation baseline models

For a comprehensive assessment of FLLA-Net's segmentation performance, we compare it with various representative semantic segmentation networks, classified according to their structural and functional characteristics:

Encoder-decoder architectures. Several encoder-decoder networks are included as foundational baselines for pixel-wise segmentation. FCN (Long et al., 2015) pioneer end-to-end semantic segmentation using a simple encoder-decoder structure. U-Net (Ronneberger et al., 2015) and U-Net++ (Zhou et al., 2019) leverage skip connections and nested dense connections to preserve spatial details and enhance gradient flow. SegNet (Badrinarayanan et al., 2017) employs pooling indices to improve decoding efficiency, while AttU-Net (Oktay et al., 2018) incorporates attention gates to focus on relevant regions. R2U-Net (AloM et al., 2018) further introduces recurrent residual learning to capture contextual information across iterations.

Multi-scale and context-aware networks. To capture long-range dependencies and global context, we include several multi-scale feature networks. The DeepLab series (DeepLabV1 (Chen, 2014), DeepLabV2 (Chen et al., 2017), DeepLabV3 (Chen, 2017), and DeepLabV3+ (Chen et al., 2018)) utilizes atrous convolution and atrous spatial pyramid pooling (ASPP) to aggregate multi-scale features. PSPNet (Zhao et al., 2017) leverages pyramid pooling to integrate global context, and RefineNet (Lin et al., 2017) fuses multi-level features for refined segmentation outputs.

Transformer-based networks. To evaluate FLLA-Net against models that leverage transformer mechanisms for capturing long-range dependencies, we focus on representative architectures. DDRNet (Hong et al., 2021) employs dual-resolution streams to capture detailed spatial and contextual information. Segformer (Xie et al., 2021) combines lightweight transformer encoders with hierarchical multi-scale feature fusion. PIDNet (Xu et al., 2023) integrates parallel interaction and detail-preserving modules to improve boundary delineation and segmentation accuracy.

3. Results

3.1. Performance comparison with other classification models

To validate the effectiveness of the proposed focused leaf-lesion awareness via explicit relational modeling (FLARE) for disease severity assessment, we first pre-train the focused leaf-lesion awareness network (FLLA-Net) on a lesion and leaf segmentation dataset. Then, we train the explicit relational modeling assessment network (ERMA-Net) using both the disease severity assessment dataset and the corresponding image descriptions, while the testing is conducted solely on the severity assessment data. We compare its performance with 15 existing

Table 3

Comparative performance of the proposed WRMA-Net framework versus 15 classification models on the disease severity assessment dataset. Results are reported as overall average accuracy and per-severity-level accuracy.

Model	mAP	Level 1	Level 3	Level 5	Level 7	Level 9
ResNet18 (He et al., 2016)	86.26	93.88	71.65	32.29	60.00	78.95
ResNet34 (He et al., 2016)	84.81	95.32	75.59	39.44	55.56	78.95
ShuffleNet (Ma et al., 2018)	82.59	94.42	59.84	47.89	55.56	82.46
DenseNet169 (Steen et al., 2018)	83.41	96.04	61.42	35.21	57.78	82.46
DenseNet264 (Steen et al., 2018)	83.29	93.53	64.57	46.48	53.33	82.41
RegNet (Radosavovic et al., 2020)	82.24	96.58	60.63	30.99	55.56	75.44
ConvNeXt (Liu et al., 2022)	79.44	93.53	55.91	35.21	57.78	66.67
EfficientNet (Tan, 2019)	85.05	97.30	69.29	39.44	62.22	75.44
EfficientNetV2 (Tan and Le, 2021)	81.08	94.06	64.57	28.17	48.89	82.46
MobileNetV2 (Sandler et al., 2018)	80.96	95.14	52.76	56.34	26.67	78.95
MobileNetV3 (Howard et al., 2019)	79.09	95.14	45.67	38.03	33.33	84.21
ViT (Dosovitskiy et al., 2020)	65.65	98.02	2.36	0.00	0.00	24.56
RepViT (Wang et al., 2024)	79.32	97.84	32.81	45.71	39.13	89.29
SBCFormer (Lu et al., 2024)	79.09	96.04	45.67	39.44	40.00	68.42
PDiscoNet (van der Klis et al., 2023)	73.95	95.68	75.59	7.04	0.00	0.00
ERMA-Net (Ours)	89.60	97.30	76.38	69.01	62.22	89.47

classification models, evaluating both the overall average accuracy and the accuracy at individual severity levels.

The evaluation results, summarized in Table 3, highlight the following key findings. First, in terms of average accuracy, FLARE significantly outperforms all baseline models, achieving at least a 3.34% improvement. This superior performance demonstrates FLARE's ability to explicitly distinguish between lesion and leaf regions—mimicking human visual awareness, and to model the relationship between their features. This design not only suppresses background interference but also simulates the spatial distribution of lesions on leaves, resulting in more accurate and human-consistent severity assessments. Second, across all evaluated severity levels, except in the first severity level, FLARE consistently achieves higher assessment accuracy compared to other models, indicating its superior ability to capture subtle differences between severity grades. This enhanced performance stems from explicitly modeling the relationship between lesions and leaves, which enables the model to perceive lesion characteristics in the context of the overall leaf condition. As a result, the model exhibits improved sensitivity and discrimination in severity grading. Finally, in contrast, while ViT achieved the highest accuracy (98.02%) in the first severity level, it shows very limited recognition performance across other levels: only 2.36% for level three, 24.56% for level nine, and even 0 for levels five and seven. This indicates that ViT suffers from a “single-level bias” problem in its evaluation, lacking the ability to distinguish between different severity levels.

3.2. Evaluation of variation in the FLLA-Net

We conduct ablation studies on the key components of FLLA-Net and perform detailed comparative analyses to evaluate the contribution of each module to lesion and leaf representation learning. FLLA-Net is specifically designed to support this learning paradigm and includes a pretrained vision awareness extraction module (VAE-Module) that explicitly extracts separate representations for lesions and leaves. Therefore, we focus our analysis on the ablation of core components of the VAE-Module, including the fast Fourier attention module (FFA-Module), the assist space constraint branch, and the effect of the VAE-Module pretraining strategy.

The results are shown in Table 4, and the key findings are as follows: First, when the VAE-Module is not pretrained, the model achieves an accuracy of only 82.26%, indicating that without leveraging the VAE-Module's ability to explicitly distinguish between lesions and leaves, the model fails to learn meaningful representations, which significantly degrades performance. Second, with a pretrained VAE-Module but without the FFA-Module and the assist space constraint branch, the accuracy is 82.52%. This demonstrates that pretraining

offers some benefit, but the lack of spatial constraints and attention mechanisms leads to imprecise feature representation and inaccurate leaf localization. Third, adding the FFA-Module under the same pre-training condition improves the accuracy to 86.67%, suggesting that incorporating frequency-based details helps representation learning. In contrast, adding only the assist space constraint branch results in a higher accuracy of 88.55%, indicating that spatial constraints play a more critical role in guiding lesion localization. Finally, when both the FFA-Module and the assist space constraint branch are included along with the pretrained VAE-Module, the model achieves the highest accuracy of 89.60%. This confirms that both fine-grained frequency information and spatial constraints are essential for accurately capturing lesion characteristics and improving assessment performance.

3.3. Evaluation of variation in the ERMA-Net

We conduct experiments on different combinations of key components in the network to analyze the impact of each module in ERMA-Net on plant leaf disease severity assessment. Specifically, first, we examine the effect of different relational modeling strategies, namely the vision relational modeling module (VRM-Module) and a simple concatenation (Concat) approach. The VRM-Module models the relationship via attention mechanisms in our method, while the concat simply fuses the lesion and leaf features along the channel dimension. For each strategy, we test three feature modeling schemes: Detail-Focused (DF) Modeling, which captures fine-grained lesion and leaf information; Context-Enhanced (CE) Modeling, which incorporates both detailed features and contextual information around the lesion; and Hierarchical-Combination (HC) Modeling, which integrates lesion and leaf details along with multi-level contextual cues. Second, since reinforcement-guided contrastive learning is introduced in our method, we evaluate the impact of different weight ratios between the main classification loss and the contrastive loss. Third, we conduct a comparative analysis between the standard contrastive loss and the contrastive loss enhanced with reinforcement-based guidance. Finally, we analyze how varying the depth of the pretrained VAE-Module influences feature extraction capacity and downstream assessment performance. The experimental results are summarized in Table 5. From the analysis, we draw the following observations:

(1) **Comparison under Different Modeling Methods.** Across all modeling schemes, using the Concat strategy results in lower accuracy compared to the proposed VRM-Module, indicating the superiority of attention-based relational modeling over direct feature concatenation. Furthermore, whether using VRM-Module or Concat, accuracy is lowest when only detailed information is used (DF). As contextual information is gradually added (from DF to CE to HC), the accuracy improves consistently. This suggests that incorporating context significantly enhances the model's ability to distinguish between severity levels.

(2) **Comparison under Different Loss Weight Settings.** When the contrastive loss is removed ($\alpha = 0$), the accuracy drops to 89.02%, highlighting the effectiveness of contrastive learning. This demonstrates that textual descriptions, serving as high-level semantic priors, can guide the model to better align visual features with corresponding severity semantics. The best performance (89.60%) is achieved when $\alpha = 0.1$ and $b = 1$, indicating that a moderate weight allows the model to benefit from text-guided supervision without overwhelming the primary assessment objective.

(3) **Comparison under Different Loss Weight Settings of the Standard Loss Function.** The reinforcement-guided contrastive learning strategy consistently outperforms the standard contrastive loss. This indicates that confidence-aware guidance enables more effective semantic alignment between visual features and severity-level descriptions, thereby improving discrimination among closely related severity levels.

(4) **Comparison under Different Depths of VAE-Module.** When the depth of the VAE-Module is set to 1, the accuracy is 84.58%. As

Table 4

Evaluation of variation in the FLLA-Net. The symbol × denotes the exclusion of the respective module from the analysis, while the checkmark symbol ✓ signifies its inclusion.

Assist space constraint branch	FFA-Module	Pre-Train VAE-Module	Accuracy
✓	✓	×	82.26
×	×	✓	82.52
×	✓	✓	86.67
✓	×	✓	88.55
✓	✓	✓	89.60

Table 5

Ablation study of key components in the ERMA-Net. We evaluate the impact of different relational modeling strategies (VRM-Module vs. simple concatenation), feature modeling schemes (Detail-Focused, Context-Enhanced, Hierarchical-Combination), contrastive loss weight ratios, and the depth of the pretrained VAE-Module on plant leaf disease severity assessment performance.

Comparison under Different Modeling Methods												
Method	VRM-Module						Concat					
	DF Modeling		CE Modeling		HC Modeling		DF Modeling		CE Modeling		HC Modeling	
Accuracy	88.79		89.02		89.60		87.62		88.20		88.90	
Comparison under Different Loss Weight Settings												
a	0.1	1	0.5	1	1	1	1.5	1	2	1	5	0
b	1	0.1	1	0.5	1	1.5	1	2	1	5	1	1
Accuracy	89.60	89.02	88.79	89.25	88.90	88.79	89.25	88.79	89.14	88.90	88.67	89.02
Comparison under Different Loss Weight Settings of the Standard Loss Function												
a	0.1	1	0.5	1	1	1	1.5	1	2	1	5	0
b	1	0.1	1	0.5	1	1.5	1	2	1	5	1	1
Accuracy	88.79	87.62	88.79	88.90	88.43	88.79	86.92	89.14	88.43	88.32	86.68	89.02
Comparison under Different Depths of VAE-Module												
Network Depth	1 Layer			2 Layers			3 Layers			4 Layers		
Accuracy	84.58			86.45			88.43			89.60		

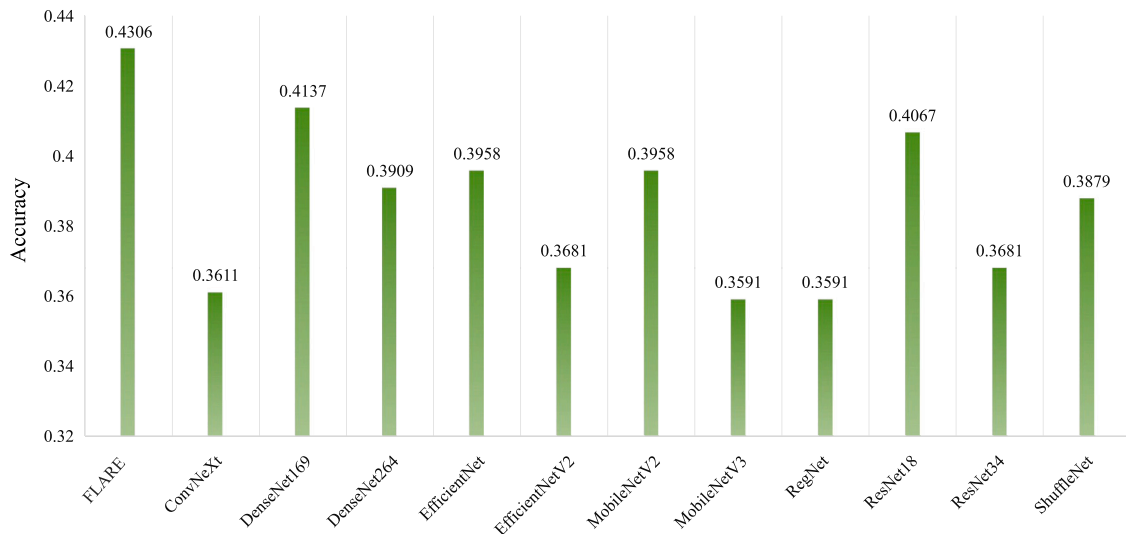


Fig. 6. Evaluation of the models' ability to adapt from complex training environments to simple testing scenarios. A total of 12 models are compared, and the proposed method achieves the best performance, demonstrating strong cross-scene robustness.

the depth increases, performance improves accordingly, demonstrating that deeper VAE-Modules enhance feature representation. However, at a depth of 3, the accuracy is 89.60%, a 1.17% decrease from the highest point, yet still outperforms all baseline classification models. This indicates that even with limited computational resources, a three-layer network can achieve competitive performance.

3.4. Verification of cross-scenario robustness

In practical applications, models are often required to adapt to varying environmental conditions, making cross-scenario robustness

a critical property. To evaluate the robustness of features learned in complex environments, we assess the assessment performance of the proposed FLARE method under simplified test scenarios. We compare FLARE with 12 models known for their high assessment accuracy. As presented in Fig. 6, FLARE achieved an accuracy of 43.06% in the simple scenario setting, outperforming all competing methods by a margin of at least 1.69 percentage points. These results demonstrate that, although trained under complex conditions, FLARE retains superior performance in simplified environments. This strongly indicates its ability to effectively transfer semantic representations across different scenarios and confirms its robust generalization capacity. Notably, the absolute accuracy of FLARE in the simple scenario is lower than that in

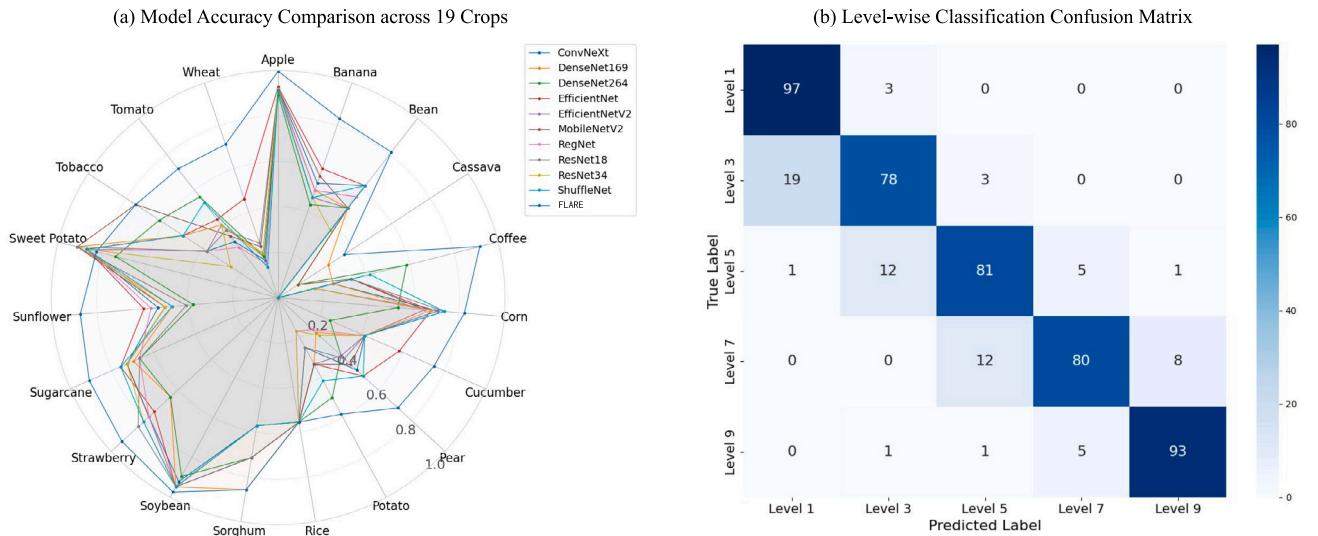


Fig. 7. Stability evaluation on species and severity. (a) Assessment accuracy on seven single-species test sets visualized using a radar chart. (b) Confusion matrix showing severity-level assessment performance.

Table 6

Experimental results comparing the segmentation performance of different models. The evaluation metrics include Precision, IoU, F1 score, and Recall for both lesion and leaf segmentation.

Model	Lesion				Leaf			
	Precision	IoU	F1 score	Recall	Precision	IoU	F1 score	Recall
FCN (Long et al., 2015)	62.62	58.59	66.97	78.12	81.57	69.70	82.01	82.72
U-Net (Ronneberger et al., 2015)	86.99	82.05	89.32	92.01	93.50	88.08	93.65	93.81
U-Net++ (Zhou et al., 2019)	93.27	88.23	94.04	94.85	97.67	95.46	97.67	97.68
AttU-Net (Oktay et al., 2018)	86.76	81.96	89.26	92.18	93.96	88.83	94.07	94.20
R2U-Net (AloM et al., 2018)	89.78	86.00	92.00	94.52	96.43	93.30	96.53	96.63
DeepLabV1 (Chen, 2014)	74.90	67.13	76.85	79.21	87.47	77.94	87.55	87.65
DeepLabV2 (Chen et al., 2017)	71.64	65.34	74.95	79.77	83.47	72.12	83.71	84.01
DeepLabV3 (Chen, 2017)	76.02	63.26	73.20	71.04	84.03	73.13	84.39	84.90
DeepLabV3+ (Chen et al., 2018)	77.85	63.43	73.48	70.51	82.51	70.33	82.49	82.48
SegNet (Badrinarayanan et al., 2017)	54.25	52.56	57.23	60.34	92.24	85.93	92.41	92.62
RefineNet (Lin et al., 2017)	70.62	66.01	74.82	78.50	95.23	90.90	95.22	95.37
PSPNet (Zhao et al., 2017)	84.06	80.58	77.69	72.22	93.84	90.29	93.91	93.98
DDRNet (Hong et al., 2021)	81.88	76.24	84.93	88.73	93.62	88.09	93.65	93.69
Segformer (Xie et al., 2021)	84.63	77.52	85.96	87.42	89.38	80.93	89.42	89.47
PIDNet (Xu et al., 2023)	88.77	83.96	90.64	92.72	96.65	93.48	96.63	96.61
FLLA-Net	93.81	89.97	94.49	95.20	97.67	95.46	97.67	97.68

the complex scenario. This is attributed to reduced contextual variability and more homogeneous visual cues, which limit the effectiveness of FLARE's relational modeling compared with complex scenarios.

3.5. Stability evaluation on species and severity levels

This experiment evaluates the assessment stability of the model on single-species test sets, as well as its performance consistency across different severity levels, which directly influences the model's reliability and practical value. First, we use a radar chart to present the assessment accuracy of 11 representative methods, including the proposed FLARE, on seventeen single-species test sets (see Fig. 7(a)). Then, we evaluate the assessment performance of the proposed FLARE method at each severity level using a confusion matrix (see Fig. 7(b)).

The results show that the proposed method achieves consistently high and stable accuracy across all single-species test sets, with the exception of Tobacco and Potato. This demonstrates the method's strong adaptability to individual species under a multi-species joint training setting. Moreover, the confusion matrix analysis further confirms the model's robustness and fine-grained recognition capability in each severity assessment. These findings validate the effectiveness of our method in delivering stable and accurate severity assessment across both species and severity levels.

3.6. Comparing FLLA-Net with other methods in image segmentation

Using lesion and leaf segmentation to pre-train FLLA-Net plays a crucial role in enhancing the explicit representation of both lesion and leaf features. To assess its segmentation performance, we evaluate the pre-trained FLLA-Net on lesion and leaf regions using standard metrics, including Precision, Intersection over Union (IoU), F1-score, and Recall. The results are compared and analyzed against 15 segmentation models.

The result is shown in Table 6, highlighting the following key findings. First, the leaf segmentation module in FLLA-Net, based on U-Net++ (Zhou et al., 2019), achieves Precision 97.67%, IoU 95.46%, F1-score 97.67%, and Recall 97.68%, outperforming all baseline models. Second, for lesion segmentation, FLLA-Net incorporates the fast Fourier attention module (FFA-module) and an assist space constraint branch into U-Net++, achieving Precision 93.81%, IoU 89.97%, F1-score 94.49%, and Recall 95.20%. These represent improvements of 0.54%, 1.74%, 0.45%, and 0.85% over the original U-Net++. FLLA-Net also surpasses other segmentation models in lesion segmentation accuracy, indicating that it has stronger capabilities in feature extraction and discrimination of lesion regions.

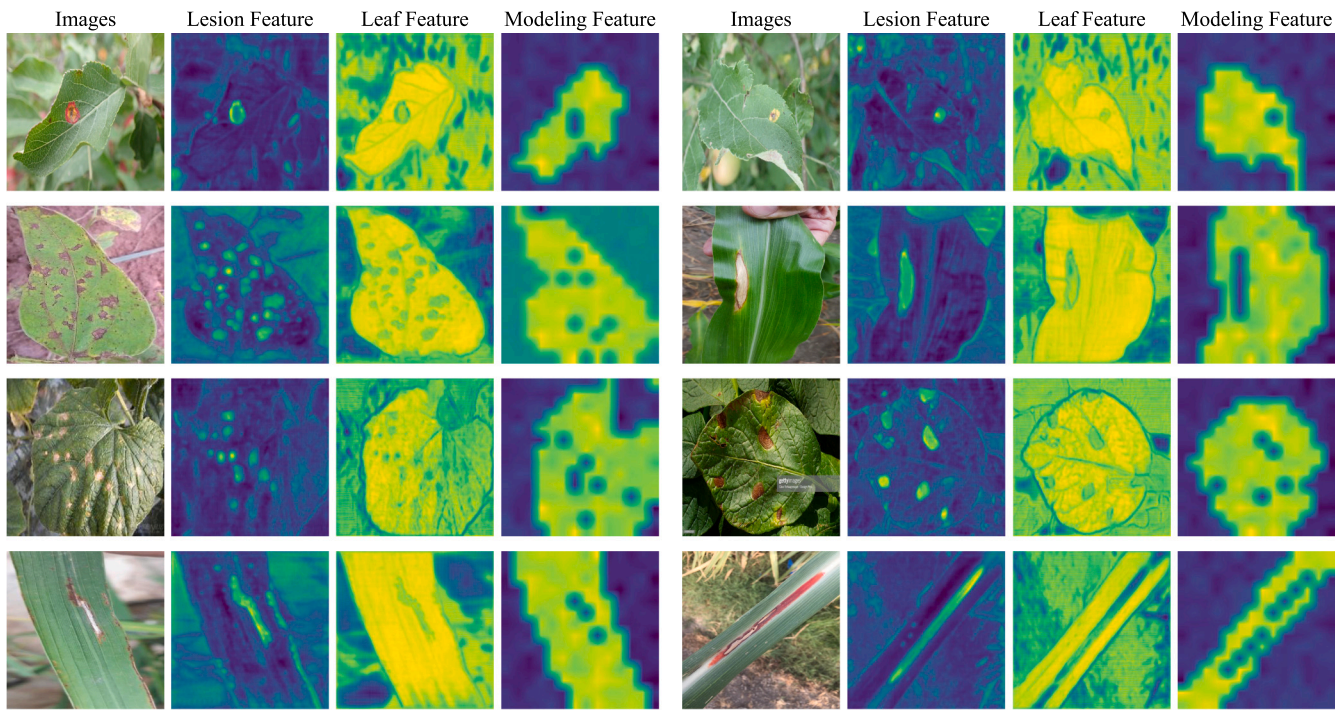


Fig. 8. Visualization of features extracted and modeled by key modules in the proposed framework. Lesion features from the lesion awareness unit, lesion-related features from the leaf awareness unit, and relational features modeled by the VRM-Module are displayed.

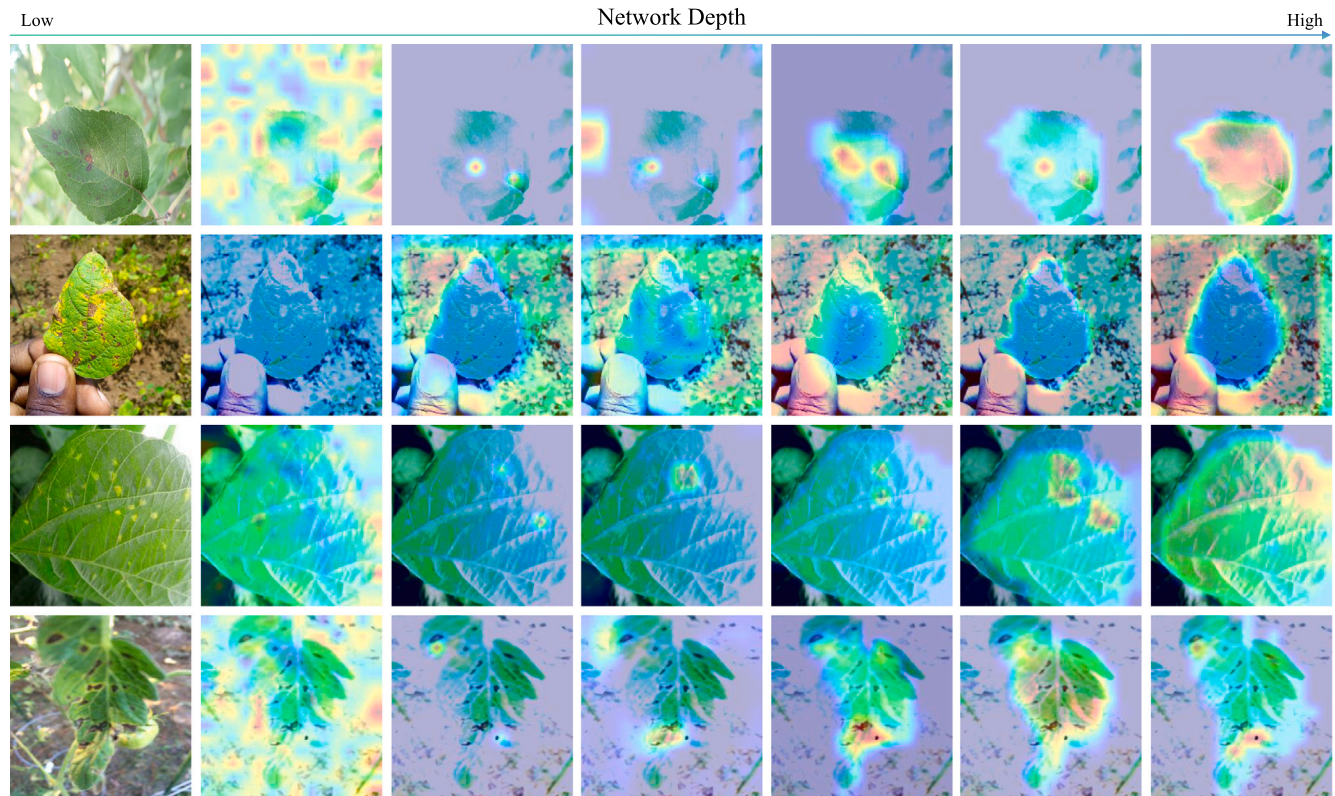


Fig. 9. Visualization of model decision-making. Grad-CAM visualizations are generated from shallow to deep layers to reveal the regions contributing most to the model's predictions.

3.7. Visualization of lesion and leaf feature

In our approach, FLLA-Net adopts a segmentation-based pretraining strategy to produce the VAE-Module, which explicitly distinguishes and

represents lesion and leaf features. ERMA-Net shares the VAE-Module for explicit vision awareness of lesions and leaves, and subsequently employs the VRM-Module to model the relationship between these features. The final severity assessment is then conducted based on the

relational features, indicating that both the extracted and modeling features are closely correlated with the severity estimation outcomes. To verify the effectiveness of the VAE-Module in extracting lesion and leaf features, as well as the relational modeling capability of the VRM-Module, we visualize the features generated by each module. Specifically, we visualize lesion features from the lesion branch of the lesion awareness unit, leaf features from the leaf awareness unit, and the relational features modeled by the VRM-Module.

The visualization results are shown in Fig. 8. It can be observed that the lesion features are focused on fine-grained lesion details, while the leaf features capture the overall structure of the leaf. After relation modeling, the modeling feature set effectively suppresses background interference and imitates the distribution of lesions on the leaf. This demonstrates that our method aligns with human visual awareness by first explicitly distinguishing lesions from leaves, followed by relational modeling, enabling more accurate and cognitively interpretable severity assessment.

3.8. Interpretability analysis of model decision-making

To further investigate the decision-making basis of the model in disease severity assessment, we employ Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017) to visualize samples from the test set. Grad-CAM works by computing the gradient of the target class with respect to the feature maps of any convolutional layer, then applying global average pooling to obtain channel-wise weights. These weights are used to generate class activation heatmaps that highlight the regions the model focuses on during prediction.

In our experiments, we visualize the class activation heatmaps across multiple network stages, from shallow to deep layers. The results, shown in Fig. 9, reveal that the model exhibits different attention patterns at different depths. First, in the shallow layers, due to limited feature extraction capability, the model's attention is relatively dispersed across the entire image without a clear focus. Second, as the network depth increases, attention gradually shifts from localized lesion regions to the broader leaf area. In the deepest layers, the model's attention consistently concentrates on the overall leaf structure while still emphasizing the lesion regions. These findings suggest that the model first attends to local lesion details, then progressively incorporates the global structural information of the entire leaf. This hierarchical attention strategy enables a more comprehensive and reasonable assessment of disease severity, aligning with the human visual cognition process and resulting in more accurate and interpretable evaluation outcomes.

3.9. Performance presentation of each severity level

To demonstrate the feature aggregation effect across different disease severity levels, we employ the t-Distributed Stochastic Neighbor Embedding (t-SNE) (Cai and Ma, 2022) method to visualize sample features. The fundamental idea of t-SNE is to preserve the distribution characteristics of the original data in a low-dimensional space. It achieves this by maintaining the relative distances between samples and their local neighborhood structures in the high-dimensional space. Specifically, we extract the intermediate feature representations from each model before the final assessment and apply t-SNE to reduce them to two dimensions for visualization of their distribution characteristics.

We select eleven models with high assessment accuracy for comparison, and their feature visualization results are presented in Fig. 10. The results indicate that the proposed method achieves superior feature aggregation, more clearly separating the five severity levels and demonstrating a stronger discriminative capability, which contributes to improved evaluation accuracy. It is worth noting that the feature aggregation of the third and fifth severity levels exhibits a certain degree of similarity in the feature space, which may be attributed to the high similarity of visual features between the two. This observation motivates us to further optimize the model to enhance its sensitivity to features at these levels, thereby improving overall performance.

4. Discussion

4.1. Segmentation for severity assessment

In plant leaf disease severity assessment, the spatial distribution and morphological characteristics of lesions serve as critical indicators for severity differentiation. However, conventional end-to-end models typically rely on hybrid perception, which often entangle lesion, leaf, and background information, thereby limiting the model's capacity to capture fine-grained semantic cues. Segmentation introduces a structured prior that guides the model to attend to semantically relevant regions, thereby enhancing the discriminative power of extracted features. Within the proposed FLARE framework, FLLA-Net adopts a dual-stream segmentation architecture to separately be aware of lesion and leaf features via the vision awareness extraction module (VAE-Module), enabling more precise localization and representation of severity-related regions. Building upon this, ERMA-Net leverages segmentation-guided feature representations and explicitly models the semantic and spatial relationship between lesions and leaves through the VRM-Module, effectively capturing the biologically meaningful indicator of lesion proportion. Compared to approaches that solely rely on global features, this relational modeling strategy facilitates more accurate severity estimation across different growth stages and lesion distributions. In summary, segmentation not only enhances the expressiveness of local lesion features but also provides a structural foundation for modeling lesion-leaf relationships, thereby significantly improving the accuracy and robustness of disease severity assessment.

4.2. Generalization discussion

The proposed FLARE framework enhances disease severity assessment by explicitly perceiving lesion and leaf features and modeling their relationships to imitate the distribution of lesions of the leaf and mimic human visual perception. Trained on complex field scenes, FLARE generalizes well to out-of-domain (OOD) categories, as shown in Fig. 11, accurately localizing features and focusing on spatial-semantic associations for precise severity prediction. This strong generalization stems from the VAE-Module, which enables FLLA-Net to learn robust, disentangled lesion-leaf representations during pretraining. These features are refined in ERMA-Net via the VRM-Module, which captures structural and contextual dependencies. Furthermore, the integration of textual semantics and reinforcement-guided contrastive learning aligns visual features with disease severity at a high semantic level. Even on unseen diseases, FLARE maintains strong feature discrimination and predictive accuracy, demonstrating its transferable multimodal knowledge and making it well-suited for real-world agricultural applications. However, differences in domain-specific terminology or missing textual annotations may limit the transferability of the semantic knowledge learned during training, because reinforcement-guided contrastive learning relies on alignment between textual and visual features. Inconsistent or absent textual information may prevent the model from learning robust associations for unseen domains, potentially affecting generalization. Nevertheless, this does not directly impact image-based inference, preserving the practical applicability of FLARE in real-world scenarios.

4.3. Advantages and disadvantages

We propose an explicit vision relational modeling (FLARE) for the intelligent assessment of plant leaf disease severity. This approach first explicitly captures the features of the lesion and leaf regions, then models their structural relationship, effectively reducing background interference and reflecting real lesion distributions. Despite these advantages, FLARE exhibits certain limitations. For example, its performance may be constrained in scenarios with limited contextual variability or highly homogeneous visual cues, where the benefits of

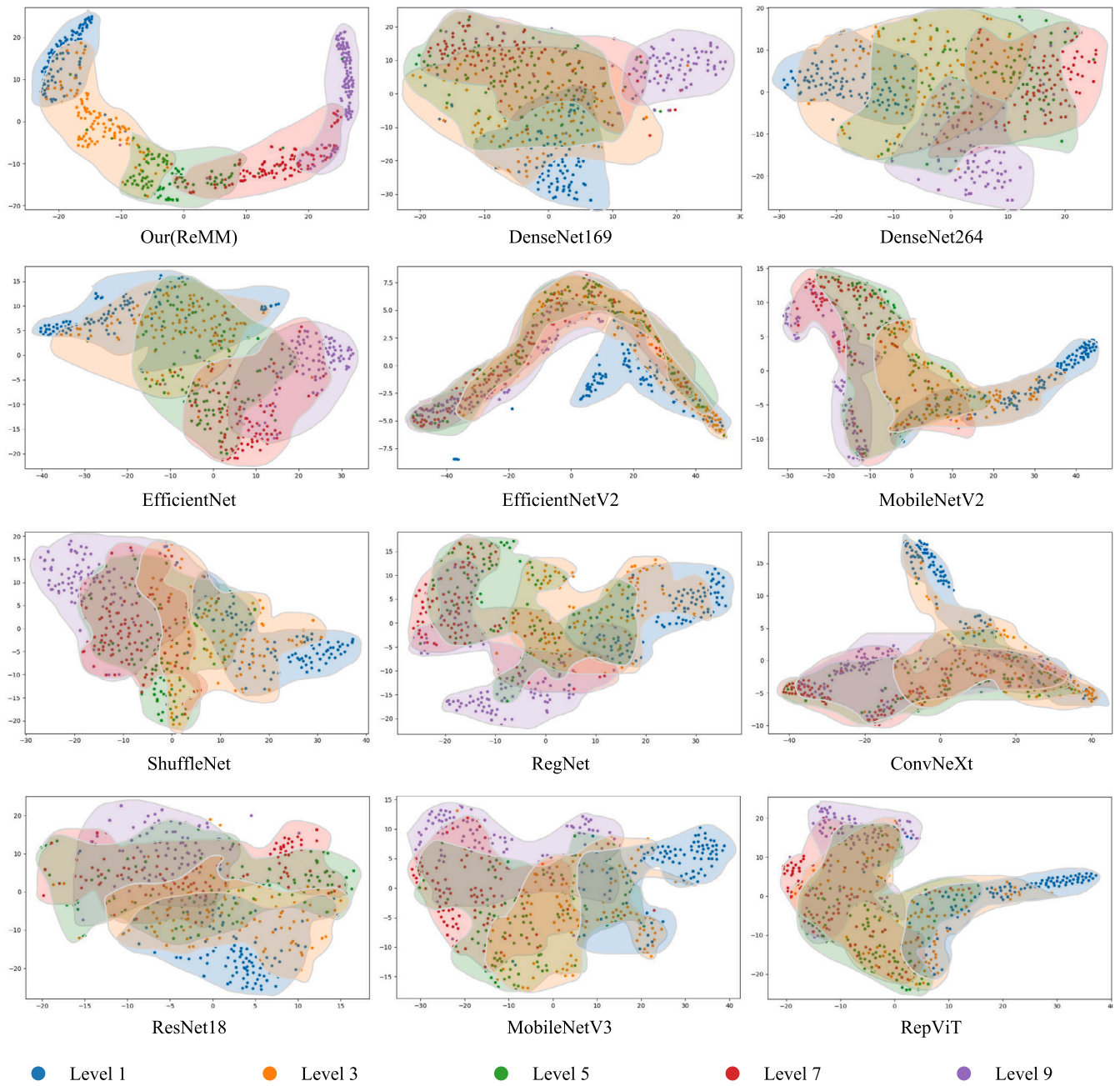


Fig. 10. t-SNE visualization of features from eleven models. The proposed method shows better separation of disease severity levels, indicating stronger feature aggregation.

relational modeling are less pronounced. In addition, limited sample availability for specific crops or severity levels may introduce data imbalance and affect generalization. The method also faces challenges in accurately localizing key regions under leaf occlusion, partial lesion coverage, or image blur, which can potentially impacting the precision of relational modeling and severity prediction.

4.4. Future prospects

The focused leaf-lesion awareness via explicit relational modeling (FLARE), developed in this study, simulates the human visual awareness process for judging disease severity. It explicitly models the semantic and spatial relationships between lesions and leaves, thereby enhancing the model's ability to distinguish between different severity levels. On the one hand, this method can be integrated into intelligent

agricultural machinery or portable detection devices to enable real-time disease severity assessment in field environments, providing efficient and scientific decision support for agricultural production. This allows farmers to apply pesticides more precisely, reduce pesticide usage, and contribute to green agriculture and sustainable development. On the other hand, when combined with drone-based inspections and IoT sensing systems, the method can be applied to remote monitoring and early warning platforms for field crops, supporting large-scale, low-cost, and high-efficiency intelligent disease monitoring and control, thereby improving the timeliness and accuracy of pest and disease management. In addition, by leveraging big data and pre-trained large models in the future, it is possible to develop generalizable severity assessment models applicable to a wide range of plant species and disease types, further promoting the advancement of intelligent plant disease assessment systems with greater generalization capability and practical value.

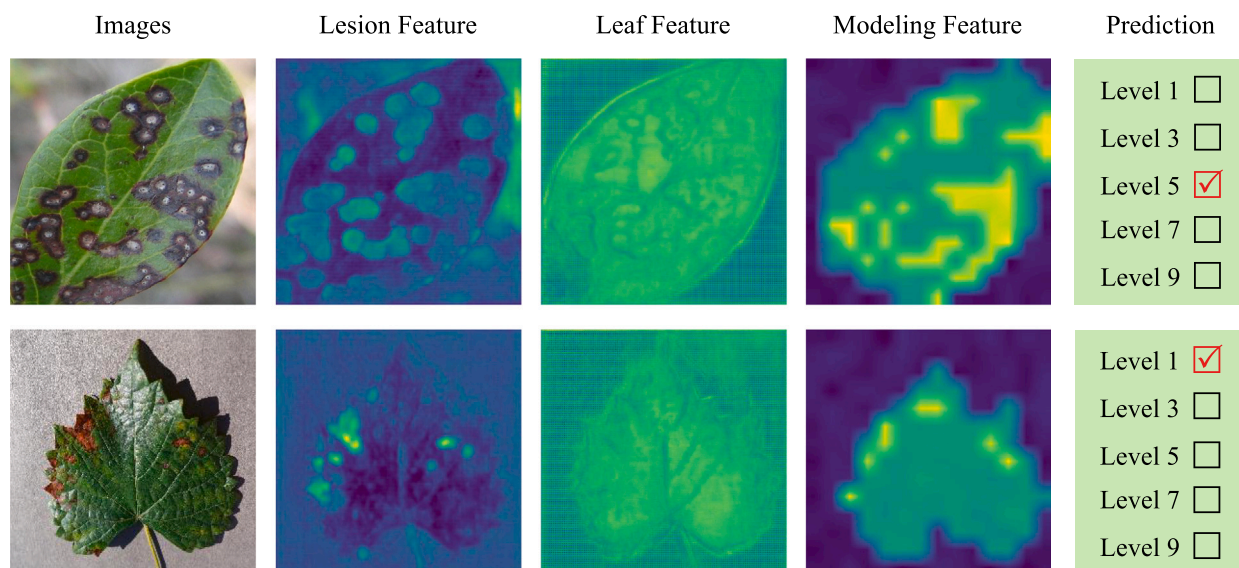


Fig. 11. Generalization evaluation of our model in unseen-category scenarios. We visualize lesion features, leaf features, relationship modeling features, and the directly predicted disease severity levels to assess generalization performance.

5. Conclusion

Accurate assessment of plant leaf disease severity is essential for guiding rational pesticide application and ensuring stable agricultural productivity. Existing methods, however, often suffer from feature entanglement among lesions, leaves, and background, which limits their robustness and accuracy in the field. To address these challenges, we propose FLARE, an explicit relational modeling of a focused leaf-lesion vision awareness framework for plant leaf disease severity assessment. The framework consists of two main components: FLLA-Net, a focused leaf-lesion awareness network that separately captures prior visual knowledge of lesions and leaves, and ERMA-Net, an explicit relational modeling assessment network that simulates the spatial distribution of lesions on leaves and predicts disease severity levels. Additionally, we construct a multi-level annotated dataset covering 19 crop species, including categories, segmentation masks, severity levels, and textual descriptions, to ensure applicability in complex real-world scenarios. Experimental results show that FLARE achieves at least a 3.34% improvement in disease severity assessment accuracy over existing methods. By suppressing background interference and explicitly modeling lesion-leaf spatial relationships, FLARE provides more accurate and vision-consistent assessments of disease severity.

CRedit authorship contribution statement

Ya Yu: Writing – original draft, methodology. **Xingcai Wu:** Writing – review & editing, Conceptualization. **Qiaoling Wan:** validation. **Yujiao Dan:** validation. **Peijia Yu:** Writing – review & editing. **Wei Guo:** Writing – review & editing. **Qi Wang:** Writing – review & editing.

Declaration of competing interest

This research was supported by Scientific and Technological Innovation Platform Research Project of Guizhou Province (CXPTXM[2025]026, CXPTXM[2025]024), the National Natural Science Foundation of China (No. 62506089), Guizhou Province Youth Science and Technology Talent Project ([2024]317), Guizhou Provincial Science and Technology Projects ([2024]002, CXTD [2023]027), Guizhou Provincial Basic Research Program (Natural Science) (Qiankehe Foundation MS[2026]080).

Data availability

The data, code, and model weights used in this paper can be downloaded from <http://flare.samlab.cn/>. In addition, this paper involves three model weights: FLLA-Net_Leaf.pth, a pre-trained leaf segmentation model; FLLA-Net_Lesion.pth, a pre-trained lesion segmentation model; and ERMA-Net.pth, a trained severity assessment network. The inference procedures for these models are described on the website.

References

- AI Challenger, 2018. AI challenger 2018 dataset. https://github.com/AIChallenger/AI_Challenger_2018.
- AloM, Md Zahangir, Hasan, Mahmudul, Yakopcic, Chris, Taha, Tarek M, Asari, Vijayan K, 2018. Recurrent residual convolutional neural network based on U-net (R2U-net) for medical image segmentation. arXiv preprint [arXiv:1802.06955](https://arxiv.org/abs/1802.06955).
- Badrinarayanan, Vijay, Kendall, Alex, Cipolla, Roberto, 2017. Segnet: A deep convolutional encoder-decoder architecture for image segmentation. *IEEE Trans. Pattern Anal. Mach. Intell.* 39 (12), 2481–2495.
- Bleasdale, Alexander J., Whyatt, J. Duncan, 2025. Classifying early apple scab infections in multispectral imagery using convolutional neural networks. *Artif. Intell. Agric.* 15 (1), 39–51.
- Bock, C.H., Poole, G.H., Parker, P.E., Gottwald, T.R., 2010. Plant disease severity estimated visually, by digital photography and image analysis, and by hyperspectral imaging. *Crit. Rev. Plant Sci.* 29 (2), 59–107.
- Cai, T. Tony, Ma, Rong, 2022. Theoretical foundations of t-sne for visualizing high-dimensional clustered data. *J. Mach. Learn. Res.* 23 (301), 1–54.
- Chen, Liang-Chieh, 2014. Semantic image segmentation with deep convolutional nets and fully connected CRFs. arXiv preprint [arXiv:1412.7062](https://arxiv.org/abs/1412.7062).
- Chen, Liang-Chieh, 2017. Rethinking atrous convolution for semantic image segmentation. arXiv preprint [arXiv:1706.05587](https://arxiv.org/abs/1706.05587).
- Chen, Liang-Chieh, Papandreou, George, Kokkinos, Iasonas, Murphy, Kevin, Yuille, Alan L., 2017. Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. *IEEE Trans. Pattern Anal. Mach. Intell.* 40 (4), 834–848.
- Chen, Liang-Chieh, Zhu, Yukun, Papandreou, George, Schroff, Florian, Adam, Hartwig, 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation. In: *Proceedings of the European Conference on Computer Vision. ECCV*, pp. 801–818.
- Dong, Xinyu, Wang, Qi, Huang, Qinding, Ge, Qinglong, Zhao, Kejun, Wu, Xingcai, Wu, Xue, Lei, Liang, Hao, Gefei, 2023. PDDD-PreTrain: a series of commonly used pre-trained models support image-based plant disease diagnosis. *Plant Phenomics* 5, 0054.
- Dosovitskiy, Alexey, Beyer, Lucas, Kolesnikov, Alexander, Weissenborn, Dirk, Zhai, Xiaohua, Unterthiner, Thomas, Dehghani, Mostafa, Minderer, Matthias, Heigold, Georg, Gelly, Sylvain, et al., 2020. An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint [arXiv:2010.11929](https://arxiv.org/abs/2010.11929).

- Gai, Yunpeng, Wang, Hongkai, 2024. Plant disease: a growing threat to global food security. *Agronomy* 14 (8), 1615.
- Gao, Chunfeng, Guo, Wei, Yang, Chenghai, Gong, Zheng, Yue, Jibo, Fu, Yuanyuan, Feng, Haikuan, 2024. A fast and lightweight detection model for wheat fusarium head blight spikes in natural environments. *Comput. Electron. Agric.* 216, 108484.
- Goyal, Lakshay, Sharma, Chandra Mani, Singh, Anupam, Singh, Pradeep Kumar, 2021. Leaf and spike wheat disease detection & classification using an improved deep convolutional architecture. *Informatics Med. Unlocked*.
- Haque, Md Ashraful, Marwaha, Sudeep, Arora, Alka, Deb, Chandan Kumar, Misra, Tanuj, Nigam, Sapna, Hooda, Karambir Singh, 2022. A lightweight convolutional neural network for recognition of severity stages of maydis leaf blight disease of maize. *Front. Plant Sci.* 13, 1077568.
- He, Kaiming, Zhang, Xiangyu, Ren, Shaoqing, Sun, Jian, 2016. Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 770–778.
- Hong, Yuanduo, Pan, Huihui, Sun, Weichao, Jia, Yisong, 2021. Deep dual-resolution networks for real-time and accurate semantic segmentation of road scenes. *arXiv preprint arXiv:2101.06085*.
- Howard, Andrew, Sandler, Mark, Chu, Grace, Chen, Liang-Chieh, Chen, Bo, Tan, Mingxing, Wang, Weijun, Zhu, Yukun, Pang, Ruoming, Vasudevan, Vijay, et al., 2019. Searching for mobilenetv3. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 1314–1324.
- Hughes, David, Salathé, Marcel, et al., 2015. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv preprint arXiv:1511.08060*.
- Kirillov, Alexander, Mintun, Eric, Ravi, Nikhila, Mao, Hanzi, Rolland, Chloe, Gustafson, Laura, Xiao, Tete, Whitehead, Spencer, Berg, Alexander C., Lo, Wan-Yen, et al., 2023. Segment anything. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 4015–4026.
- van der Klis, Robert, Alaniz, Stephan, Mancini, Massimiliano, Dantas, Cassio F, Ienco, Dino, Akata, Zeynep, Marcos, Diego, 2023. PDiscoNet: Semantically consistent part discovery for fine-grained recognition. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 1866–1876.
- Kumari, Karishma, Paray, Roaf, Basavaraj, Y.B., Godara, Samarth, Mani, Indra, Kumar, Rajeev, Khura, Tapan, Sarkar, Susheel, Ranjan, Rajeev, Mirzakhani-nafchi, Hasan, 2025. Spectral sensor-based device for real-time detection and severity estimation of groundnut bud necrosis virus in tomato. *J. Field Robot.* 42 (1), 5–19.
- Leite, Douglas, Brito, Alisson, Faccioli, Gregorio, 2024. Advancements and outlooks in utilizing convolutional neural networks for plant disease severity assessment: A comprehensive review. *Smart Agric. Technol.* 9, 100573.
- Lin, Guosheng, Milan, Anton, Shen, Chunhua, Reid, Ian, 2017. Refinenet: Multi-path refinement networks for high-resolution semantic segmentation. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 1925–1934.
- Liu, Xiang, Liu, Zhaoxiang, Hu, Huan, Chen, Zezhou, Wang, Kohou, Wang, Kai, Lian, Shiguo, 2024. A multimodal benchmark dataset and model for crop disease diagnosis. In: *European Conference on Computer Vision*. Springer, pp. 157–170.
- Liu, Zhuang, Mao, Hanzi, Wu, Chao-Yuan, Feichtenhofer, Christoph, Darrell, Trevor, Xie, Saining, 2022. A convnet for the 2020s. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 11976–11986.
- Long, Jonathan, Shelhamer, Evan, Darrell, Trevor, 2015. Fully convolutional networks for semantic segmentation. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. CVPR.
- Lu, Xiangyong, Suganuma, Masanori, Okatani, Takayuki, 2024. Sbcformer: Lightweight network capable of full-size ImageNet classification at 1 FPS on single board computers. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. pp. 1123–1133.
- Ma, Ningning, Zhang, Xiangyu, Zheng, Hai-Tao, Sun, Jian, 2018. Shufflenet v2: Practical guidelines for efficient cnn architecture design. In: *Proceedings of the European Conference on Computer Vision*. ECCV, pp. 116–131.
- Mahlein, Anne-Katrin, 2016. Plant disease detection by imaging sensors—parallels and specific demands for precision agriculture and plant phenotyping. *Plant Dis.* 100 (2), 241–251.
- Mehra, Lucky K, Cowger, Christina, Gross, Kevin, Ojiambo, Peter S, 2016. Predicting pre-planting risk of stagonospora nodorum blotch in winter wheat using machine learning models. *Front. Plant Sci.* 7, 390.
- National public service platform for standards information, 0000. NY/T 992-2006 Air-assisted orchard sprayer - Operation quality, Technical report, Accessed: <https://std.samr.gov.cn/hb/search/stdHBDetailed?id=B82386E15E5A7214E05397BE0A0AE9BD>.
- Oktay, Ozan, Schlemper, Jonathan, Folgoc, Luan, Lee, Meng-Hao, Heinrich, Matthew P, Misawa, Takeru, Mori, Koichi, Hammer, Gabriel, Glocker, Ben, Rueckert, Daniel, 2018. Attention U-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*.
- Omaye, James Daniel, Ogbuju, Emeka, Ataguba, Grace, Jaiyeoba, Oluwayemisi, Aneke, Joseph, Oladipo, Francisca, 2024. Cross-comparative review of machine learning for plant disease detection: Apple, cassava, cotton and potato plants. *Artif. Intell. Agric.*.
- Pacal, Ishak, Kunderacioglu, Ismail, Alma, Mehmet Hakki, Deveci, Muhammet, Kadry, Seifedine, Nedoma, Jan, Slany, Vlastimil, Martinek, Radek, 2024. A systematic review of deep learning techniques for plant diseases. *Artif. Intell. Rev.* 57 (11), 304.
- Radford, Alec, Kim, Jong Wook, Hallacy, Chris, Ramesh, Aditya, Goh, Gabriel, Agarwal, Sandhini, Sastry, Girish, Askell, Amanda, Mishkin, Pamela, Clark, Jack, et al., 2021. Learning transferable visual models from natural language supervision. In: *International Conference on Machine Learning*. PMLR, pp. 8748–8763.
- Radosavovic, Ilija, Kosaraju, Raj Prateek, Girshick, Ross, He, Kaiming, Dollár, Piotr, 2020. Designing network design spaces. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 10428–10436.
- Ronneberger, Olaf, Fischer, Philipp, Brox, Thomas, 2015. U-net: Convolutional networks for biomedical image segmentation. In: *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III* 18. Springer, pp. 234–241.
- Salamai, Abdullah Ali, 2024. Towards automated, efficient, and interpretable diagnosis coffee leaf disease: A dual-path visual transformer network. *Expert Syst. Appl.* 255, 124490.
- Sandler, Mark, Howard, Andrew, Zhu, Menglong, Zhmoginov, Andrey, Chen, Liang-Chieh, 2018. Mobilenetv2: Inverted residuals and linear bottlenecks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 4510–4520.
- Selvaraju, Ramprasaath R., Cogswell, Michael, Das, Abhishek, Vedantam, Ramakrishna, Parikh, Devi, Batra, Dhruv, 2017. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: *2017 IEEE International Conference on Computer Vision*. ICCV, pp. 618–626.
- Singh, R.N., Krishnan, P., Bharadwaj, C., Das, Bappa, 2023. Improving prediction of chickpea wilt severity using machine learning coupled with model combination techniques under field conditions. *Ecol. Informatics* 73, 101933.
- Sladojevic, Srdjan, Arsenovic, Marko, Anderla, Andras, Culibrk, Dubravko, Stefanovic, Darko, 2016. Deep neural networks based recognition of plant diseases by leaf image classification. *Comput. Intell. Neurosci.* 2016 (1), 3289801.
- Steen, M., Downe, S., Bamford, N., Edozien, L., 2018. DenseNet: densely connected convolutional networks. pp. 362–371, *arXiv preprint arXiv:1608.06993*.
- Steininger, Daniel, Trondl, Andreas, Croonen, Gerardus, Simon, Julia, Widhalm, Verena, 2023. The cropandweed dataset: A multi-modal learning approach for efficient crop and weed manipulation. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. pp. 3729–3738.
- Tan, Mingxing, 2019. Efficientnet: Rethinking model scaling for convolutional neural networks. *arXiv preprint arXiv:1905.11946*.
- Tan, Mingxing, Le, Quoc, 2021. Efficientnetv2: Smaller models and faster training. In: *International Conference on Machine Learning*. PMLR, pp. 10096–10106.
- Wang, Ao, Chen, Hui, Lin, Zijia, Han, Jungong, Ding, Guiguang, 2024. Repvit: Revisiting mobile cnn from vit perspective. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 15909–15920.
- Weyler, Jan, Magistri, Federico, Marks, Elias, Chong, Yue Linn, Sodano, Matteo, Roggiolani, Gianmarco, Chebrolu, Nived, Stachniss, Cyrill, Behley, Jens, 2024. PhenoBench: A large dataset and benchmarks for semantic image interpretation in the agricultural domain. *IEEE Trans. Pattern Anal. Mach. Intell.*.
- Xie, Enze, Wang, Wenhui, Yu, Zhiding, Anandkumar, Anima, Alvarez, Jose M., Luo, Ping, 2021. SegFormer: Simple and efficient design for semantic segmentation with transformers. *Adv. Neural Inf. Process. Syst.* 34, 12077–12090.
- Xu, Jiacong, Xiong, Zixiang, Bhattacharyya, Shankar P., 2023. PIDNet: A real-time semantic segmentation network inspired by pid controllers. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 19529–19539.
- Yang, Bin, Wang, Zhulian, Guo, Jinyuan, Guo, Lili, Liang, Qiaokang, Zeng, Qiu, Zhao, Ruiyuan, Wang, Jianwu, Li, Caihong, 2023. Identifying plant disease and severity from leaves: A deep multitask learning framework using triple-branch swin transformer and deep supervision. *Comput. Electron. Agric.* 209, 107809.
- Zhao, Hengshuang, Shi, Jianping, Qi, Xiaojuan, Wang, Xiaogang, Jia, Jiaya, 2017. Pyramid scene parsing network. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 2881–2890.
- Zhou, Zongwei, Siddiquee, Md Mahfuzur Rahman, Tajbakhsh, Nima, Liang, Jianming, 2019. Unet++: Redesigning skip connections to exploit multiscale features in image segmentation. *IEEE Trans. Med. Imaging* 39 (6), 1856–1867.